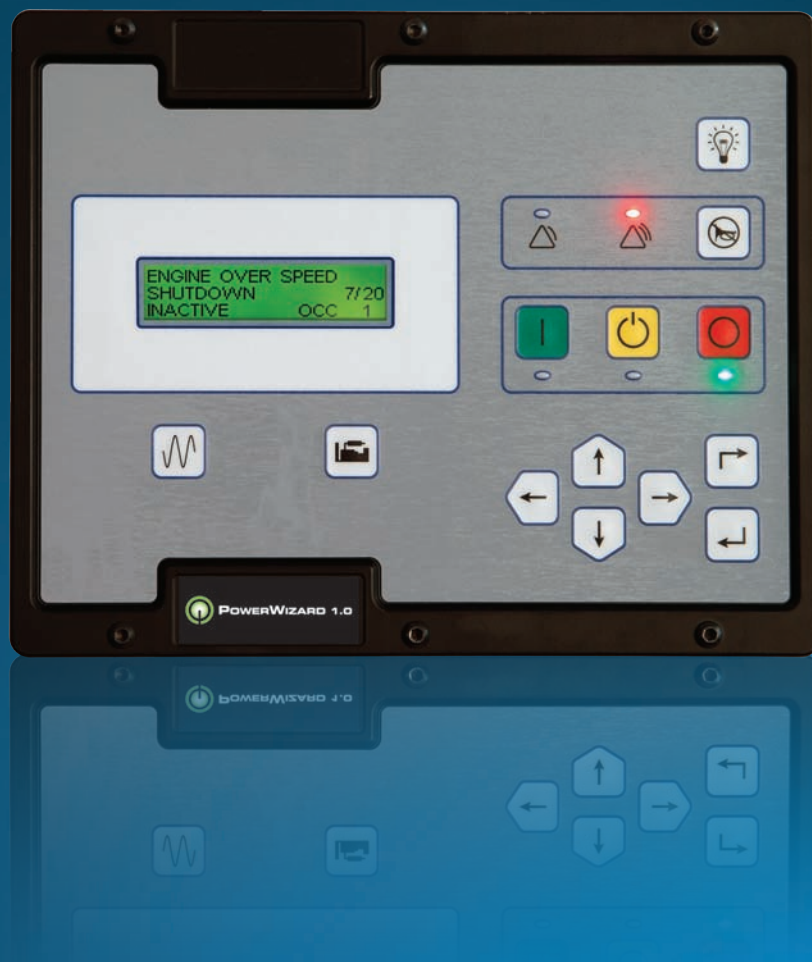


PowerWizard 1.0 & 2.0 Control Systems

Technical Manual



Important Safety Information

Most accidents that involve product operation, maintenance and repair are caused by failure to observe basic safety rules or precautions. An accident can often be avoided by recognizing potentially hazardous situations before an accident occurs.

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1 General Information

1.1 Introduction



Figure 1: PowerWizard Control System Panel

The controller is available in two versions, PowerWizard 1.0 and PowerWizard 2.0. These two versions are based on different features.

This Application and Installation Guide is intended to cover the PowerWizard Generating Set Control and its application in generating set systems. The intended audience for this guide includes generating set system designers, service support personnel, Dealers and service technicians.

1.2 Applications

The PowerWizard range of generating set control systems is designed for use in a variety of applications. The control systems can be used on standby and prime power diesel generating sets. The configurability of the controllers allows them to be used, in some cases, on other applications such as marine auxiliary generating sets, switchgear applications, industrial engines and generating sets as well as gas generating sets.

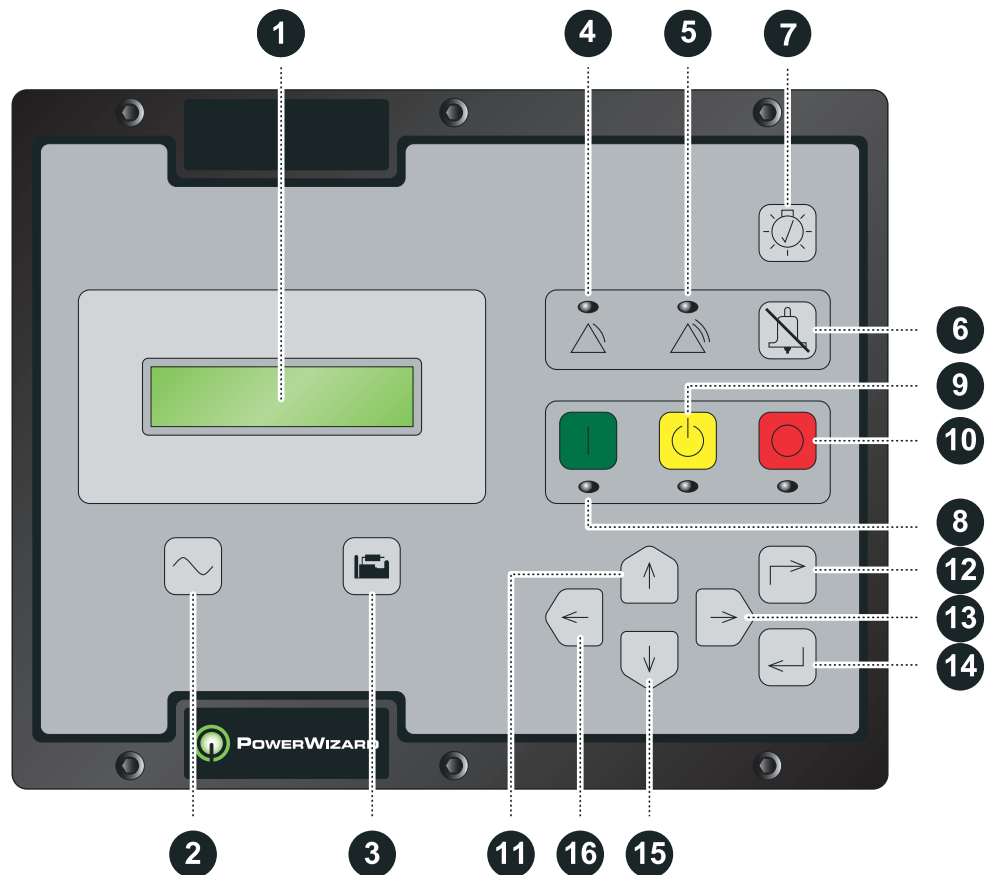
1.3 PowerWizard Variations (PW1.0 and PW2.0)

Some of the different features of the two versions, PowerWizard 1.0 and PowerWizard 2.0 are listed in Table 1.

Controllers Series Features	PowerWizard 1.0	PowerWizard 2.0
AC Volts, Current and Frequency Metering	✓	✓
AC Power Metering	-	✓
DC metering – Batt Volts, Eng Hours Run, RPM, Eng Temperature and Oil Pressure	✓	✓
Spare Analogue Inputs	0	1
Spare Input Fault Channels	2	4
Spare Outputs Volt Free Contacts	0	2
Programmable Sink Output	0	1
Real Time Clock	-	✓
20 Event Fault Log	✓	✓
2 Display Languages (customer language + Tech English)	✓	✓
Service Interval Counter	-	✓
CAN1 J1939 Data Link	✓	✓
CAN2 J1939 Data Link	-	✓
SCADA RS485 Modbus Remote Monitoring and Control	-	✓

Table 1: Features available on PowerWizard Modules

1.4 PowerWizard Control Module Description



- | | |
|------------------------------|------------------------|
| 1 Display screen | 9 Auto |
| 2 AC overview hot-key | 10 Stop |
| 3 DC overview hot-key | 11 Up cursor |
| 4 Warning lamp | 12 Escape |
| 5 Shutdown lamp | 13 Right cursor |
| 6 Alarm acknowledge | 14 Enter |
| 7 Lamp test | 15 Down cursor |
| 8 Run | 16 Left cursor |

Figure 2: PowerWizard Control Module Description

2 Basic Operation

2.1 START Mode



Press **START** Key

2.2 STOP Mode



Press **STOP** Key

2.3 AUTO Mode

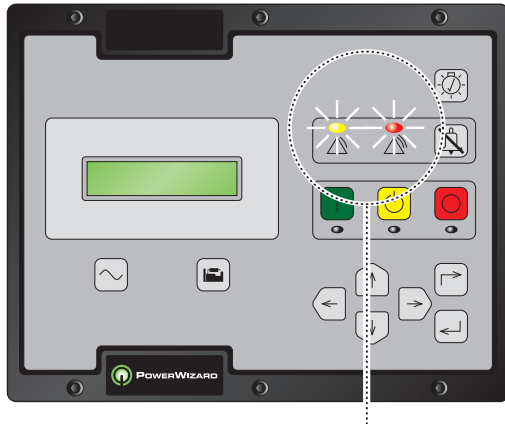


Press **AUTO** Key

Figure 3: Basic Operation – Start, Stop and Auto Modes

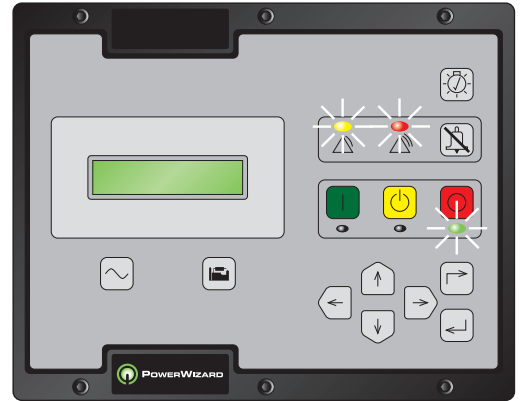
2.4 Fault / Alarm Reset Process

1 Fault / Alarm Reset Process



If either of these indication lamps are flashing or solid there is a warning or shutdown

2 Fault / Alarm Reset



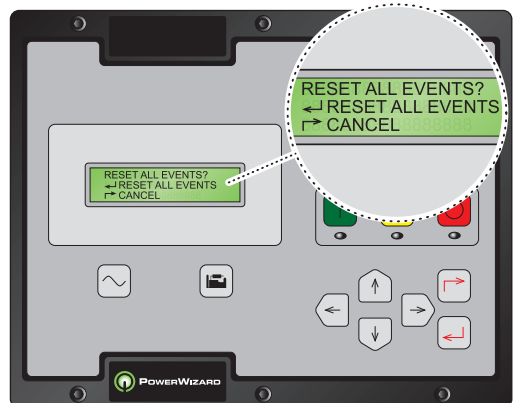
 Press **STOP** Key

3 Fault / Alarm Reset



 Press and hold **"Alarm Acknowledge"** Key for 3 seconds

4 Fault / Alarm Reset. The display will show:



 Press **ENTER** Key to clear all Warnings and / or Shutdowns

 Press **ESCAPE** Key to cancel

Figure 4: Basic Operation – Fault Alarm Reset Process

3 User Interface Overview

3.1 Function Keys

 **AC Overview hot-key** – The AC Overview key will navigate the display to the first screen of AC information. The AC Overview information contains various AC parameters that summarise the electrical operation of the generating set. (Use the up/down keys to navigate within the AC parameters).

 **Engine Overview hot-key** – The Engine Overview key will navigate the display to the first screen of engine information. The Engine Overview information contains various engine parameters that summarise the operation of the generating sets. (Use the up/down keys to navigate within the Engine parameters).

 **Lamp Test** – Pressing and holding the Lamp Test key will cause all of the LED's and the display screen pixels to turn on.

 **RUN** – Pressing the Run key will cause the engine to enter the run mode.

 **AUTO** – Pressing the Auto key will cause the engine to enter the auto mode.

 **STOP** – Pressing the Stop key will cause the engine to enter stop mode.

3.2 Menu Navigators

 **Scroll Up** – The Scroll Up key is used to navigate up through the various menus or monitoring screens. The Scroll Up key is also used during setpoint entry. During numeric data entry the Scroll Up key is used in order to increment the digits (0-9). If the setpoint requires selection from a list, the Scroll Up key is used to navigate through the list.

 **Escape** – The Escape key is used during menu navigation in order to navigate up through the menu/sub-menu structure. Each key press causes the user to move backwards/upwards through the navigation menus. The Escape key is also used to exit/cancel out of data entry screens during setpoint programming. If the Escape key is pressed during setpoint programming, none of the changes made on screen will be saved to memory.

 **Scroll Right** – The Scroll Right key is used during setpoint adjustment. During numeric data entry, the Scroll Right key is used to choose which digit is being edited. The Scroll Right key is also used during certain setpoint adjustments to select or deselect a check box. If a box has a check mark inside, pressing the Scroll Right key will cause the check mark to disappear, disabling the function. If the box does not have a check mark inside, pressing the Scroll Right key will cause a check mark to appear, enabling the function.

 **Enter** – The Enter key is used during menu navigation to select menu items in order to navigate forward/downward in the menu/sub-menu structure. The Enter key is also used during setpoint programming in order to save setpoint changes. Pressing the Enter key during setpoint programming causes setpoint changes to be saved to memory.

 **Scroll Down** – The Scroll Down key is used to navigate down through the various menus or monitoring screens. The Scroll Down key is also used during setpoint entry. During numeric data entry the Scroll Down key is used in order to decrement the digits (0-9). If the setpoint requires selection from a list, the Scroll Down key is used to navigate down through the list.

 **Scroll Left** – The Scroll Left key is used during setpoint adjustment. During numeric data entry, the Scroll Left key is used to choose which digit is being edited. The Scroll Left key is also used during certain setpoint adjustments to select or deselect a check box. If a box has a check mark inside, pressing the Scroll Left key will cause the check mark to disappear, disabling the function. If the box does not have a check mark inside, pressing the Scroll Left key will cause a check mark to appear, enabling the function.

3.3 Alarm Indicators



Yellow Warning Light – A flashing yellow light indicates that there are unacknowledged active warnings. A solid yellow light indicates that there are acknowledged warnings active. If there are any active warnings, the yellow light will change from flashing yellow to solid yellow after the Alarm Acknowledge key is pressed. If there are no longer any active warnings, the yellow light will turn off after the Alarm Acknowledge key is pressed.



Red Shutdown Light – A flashing red light indicates that there are unacknowledged active shutdown events. A solid red light indicates that there are acknowledged shutdown events active. If there are any active shutdown events the red light will change from flashing red to solid red after the Alarm Acknowledge key is pressed. Any condition that has caused a shutdown event must be manually reset. If there are no longer any active shutdown events, the red light will turn off.



Alarm Acknowledge – Pressing the Alarm Acknowledge will cause the horn relay output to turn off and silence the horn (if installed). Pressing the key will also cause any yellow or red flashing lights to turn off or to become solid depending on the active status of the alarms. The Alarm Acknowledge may also be configured to send out a global alarm silence on the J1939 Data Link, which will silence horns on annunciators. Pressing and holding the Alarm Acknowledge key can be used to reset all active warnings or shutdowns.

4 Detailed Operation

4.1 PowerWizard Menu Trees

4.1.1 PowerWizard 1.0 Menu Tree

PowerWizard 1.0 Generating Set Control Menu Structure

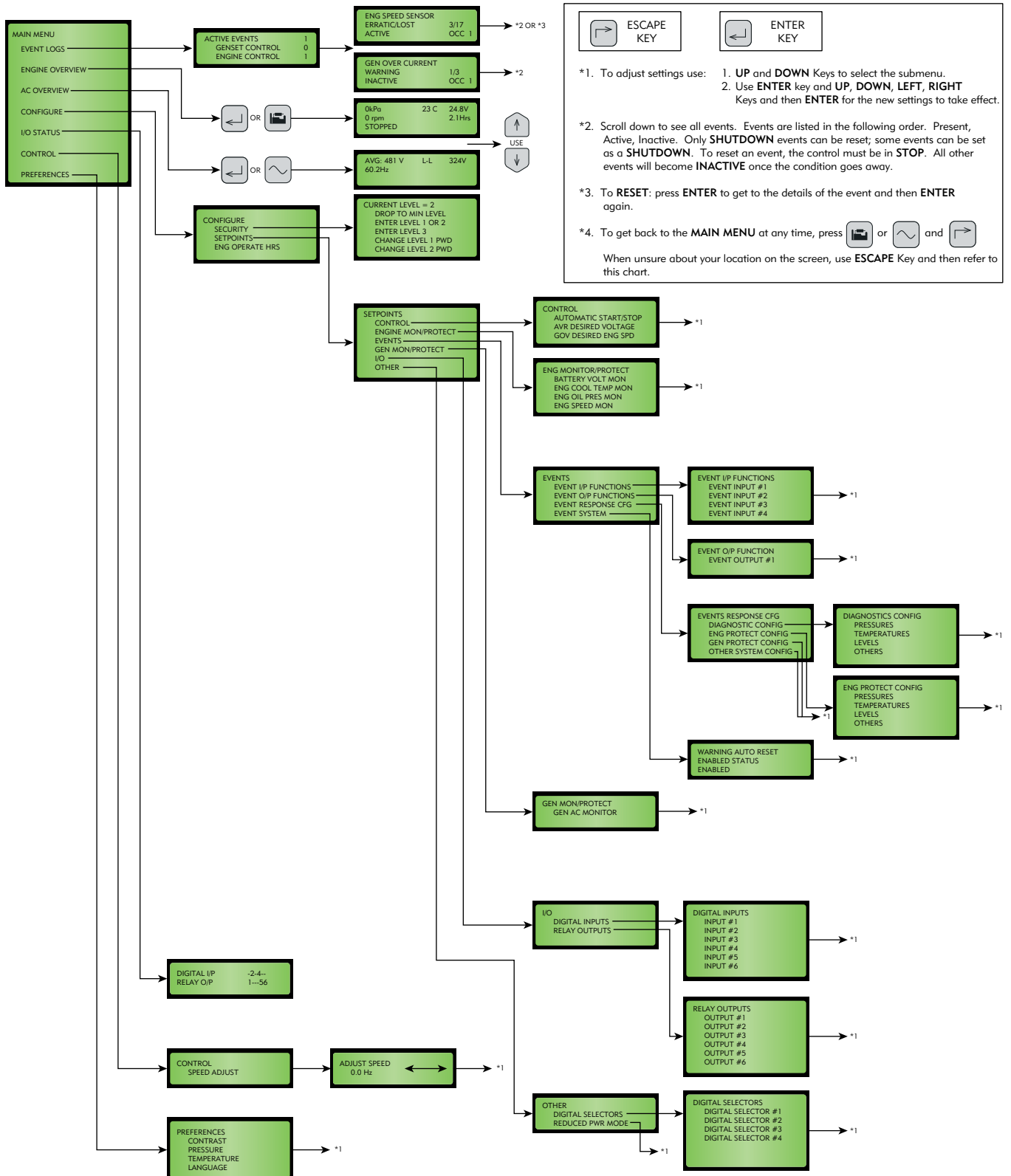


Figure 5: PowerWizard 1.0 Menu Tree

MAIN MENU

- EVENT LOG
- ENGINE OVERVIEW
- AC OVERVIEW
- CONFIGURE
- I/O STATUS
- CONTROL
- PREFERENCES

CONFIGURE

- ACTIVE EVENTS
 - GENSET CONTROL
 - ENGINE CONTROL
- SETPOINTS
 - CONTROL
 - ENGINE MON/PROTECT
 - EVENTS
 - GEN MON/PROTECT
 - I/O
 - NETWORK
 - OTHER
- CONTROL
 - DIGITAL I/P
 - RELAY O/P
 - DIGITAL O/P
- EVENTS
 - EVENT Q/P FUNCTIONS
 - EVENTS RESPONSE CFG
 - EVENTS Q/P FUNCTION
 - EVENTS RESPONSE CFG
 - EVENTS Q/P FUNCTION
 - EVENTS RESPONSE CFG
 - EVENTS Q/P FUNCTION
 - EVENTS RESPONSE CFG
 - EVENTS Q/P FUNCTION
 - EVENTS RESPONSE CFG
- I/O
 - DIGITAL INPUTS
 - INPUT #1
 - INPUT #2
 - INPUT #3
 - INPUT #4
 - INPUT #5
 - INPUT #6
 - INPUT #7
 - INPUT #8
- NETWORK
 - DATALINK - SCADA
- OTHER
 - DIGITAL SELECTORS
 - DIGITAL SELECTOR #1
 - DIGITAL SELECTOR #2
 - DIGITAL SELECTOR #3
 - DIGITAL SELECTOR #4
 - DIGITAL SELECTOR #5
 - DIGITAL SELECTOR #6
 - DIGITAL SELECTOR #7

CONTROL

- DIGITAL I/P
- RELAY O/P
- DIGITAL O/P

EVENTS

- EVENT Q/P FUNCTIONS
- EVENTS RESPONSE CFG
- EVENTS Q/P FUNCTION
- EVENTS RESPONSE CFG
- EVENTS Q/P FUNCTION
- EVENTS RESPONSE CFG
- EVENTS Q/P FUNCTION
- EVENTS RESPONSE CFG
- EVENTS Q/P FUNCTION
- EVENTS RESPONSE CFG

I/O

- DIGITAL INPUTS
- INPUT #1
- INPUT #2
- INPUT #3
- INPUT #4
- INPUT #5
- INPUT #6
- INPUT #7
- INPUT #8

NETWORK

- DATALINK - SCADA

OTHER

- DIGITAL SELECTORS
- DIGITAL SELECTOR #1
- DIGITAL SELECTOR #2
- DIGITAL SELECTOR #3
- DIGITAL SELECTOR #4
- DIGITAL SELECTOR #5
- DIGITAL SELECTOR #6
- DIGITAL SELECTOR #7

LEGEND

- ESCAPE KEY
- ENTER KEY

INSTRUCTIONS

- To adjust settings use:
 - UP and DOWN Keys to select the submenu.
 - Use ENTER Key and UP, DOWN, LEFT, RIGHT Keys and then ENTER for the new settings to take effect.
- Scroll down to see all events. Events are listed in the following order, Present, Active, Inactive. Only SHUTDOWN events can be reset; some events can be set as a SHUTDOWN. To reset an event, the control must be in STOP. All other events will become INACTIVE once the condition goes away.
- To RESET: press ENTER to get to the details of the event and then ENTER again.
- To get back to the MAIN MENU at any time, press [BACK] or [HOME] and [ENTER].

WHEN UNREASSURED ABOUT YOUR LOCATION ON THE SCREEN, USE ESCAPE KEY AND THEN REFER TO THIS CHART.

Figure 6: PowerWizard 2.0 Menu Tree

4.2 Technical Operation

4.2.1 Engine Starting Sequence

1. The PowerWizard receives an engine start signal. The signal will be one of four:
 - The operator presses the run key
 - The control is in auto and the remote initiate digital input becomes active
 - The control receives a start command via the RS-485 SCADA Data Link (PowerWizard 2.0 only)
 - The control has a program cycle timer (PCT) configured that becomes active (PowerWizard 2.0 only).
2. The PowerWizard checks the system before beginning the cranking sequence. The PowerWizard checks that there are no shutdown events present and that all previous shutdown events have been reset.
3. If the engine is equipped with a form of start aid such as thermostart it will enter a prestart sequence.
4. The PowerWizard begins the crank sequence by activating the engine crank relay and the fuel control relay:
 - If the engine reaches the setpoint for crank terminate speed the engine is deemed running and the engine crank relay will be deactivated
 - If the engine fails to start within a preset duration (normally seven seconds) the PowerWizard will stop cranking and wait for a pre-set time (normally seven seconds) before attempting to start again
 - If the PowerWizard fails to start within a pre-set number of crank attempts (normally three) the PowerWizard will activate a "Fail to Start shutdown".

4.2.2 Engine Stopping Sequence

1. The PowerWizard receives an engine stop signal. The signal will be one of four:
 - The operator presses the Stop key
 - The control is in auto and the remote initiate digital input becomes inactive
 - The control receives a stop command via the RS-485 SCADA Data Link (PowerWizard 2.0 only)
 - The control has a program cycle timer (PCT) configured that becomes inactive (PowerWizard 2.0 only).
2. The PowerWizard begins the cool down period. In order to bypass the cool down hold down the stop key for 3 seconds. The options "PRESS ENTER TO BYPASS" and "PRESS ESCAPE TO CONTINUE" will be shown on the display. Press the Enter key to bypass the cool down sequence or press the Escape key to continue the cool down sequence.
3. After the cool down cycle, the PowerWizard stops the engine by turning off the fuel control relay.

4.2.3 Event System

The PowerWizard module uses the J1939 standard format for events, whereby an event is defined as a combination of a suspect parameter number (SPN) and a failure mode identifier (FMI). The SPN defines what is at fault and the FMI defines how it is at fault (e.g. SPN = 100 = Oil Pressure, FMI = 1 = Low Shutdown would mean the set has shutdown on a low oil pressure fault). A list of SPN/FMI combinations can be found in appendix A.

PowerWizard modules have separate event logs for events raised by the module itself and those raised by ancillaries such as the engine ECM.

4.2.4 Event State

Events in PowerWizard may exist in one of three states: **present**, **active** and **inactive**.

Present – The condition causing the event is ongoing and affecting system behaviour. If an event is present it cannot be reset.

Active – The event is no longer present and can be reset by the user. However it has been latched by the event system and needs to be reset before the engine can be restarted.

Inactive – The event was active at some time but is no longer active and is not affecting system behaviour. Inactive events exist for historical purposes only.

4.2.5 Event Viewing

1. From the main menu, highlight "EVENT LOGS" and press the enter key. A list of all installed ECMs will be displayed; the number opposite represents the total of present and active events in each log.
2. Highlight an ECM and press the Enter key to view that ECMs event log.
3. In order to scroll through the events use the Up and Down keys. Events are ordered with present events first, active events next and inactive events last. Within these classifications they are ordered by engine run hours (or real time clock on PowerWizard 2.0).
4. Press enter after highlighting an event to see additional information such as SPN, FMI, time and date of first occurrence, time and date of last occurrence (PowerWizard 2.0 only), engine hours at first occurrence and engine hours at last occurrence.

4.2.6 Single Event Resetting

A flashing red shutdown light indicates there is an unacknowledged shutdown event. The red shutdown light will change from flashing red to solid red when the Alarm Acknowledged key is pressed. Once a fault has been checked and the cause rectified, use the following procedure in order to reset the event:

1. Press the Stop key.
2. Enter the "EVENT LOGS" option from the main menu.
3. Select an ECM from the list.
4. Scroll through the events in order to highlight the event to be reset.
5. Make sure the event status is active (not present).
6. Press the Enter key.
7. "RESET" will be highlighted if the condition is no longer present and the control is in Stop.
8. Press the Enter key again. The fault will clear.
9. Press the Escape key 3 times in order to get back to the main menu.

4.2.7 Quick Event Resetting

In addition to the above procedure there is also a simplified process for resetting all events. To reset all events:

1. Press the Stop key.
2. Press and hold the Alarm Acknowledge key for three seconds.
3. Press Enter to reset all events, press Escape to cancel.

Note – the PowerWizard must be in stop mode to reset events.

Note – present events cannot be reset.

4.3 Security

There are 3 levels of password protection on the PowerWizard control panel. All of the adjustable setpoints are associated with a specific level of security required to make an adjustment to the parameter. The passwords only affect changing setpoints within the control panel. Changing setpoints with the Service Tool does not require passwords.

The level of password protection that is required for each setpoint is identified on the parameter setpoint entry screen. A security level identification number "1", "2" or "3" next to a padlock symbol is displayed on the parameter setpoint entry screen. A Level 3 security is used for the most secure setpoints and Level 1 security is used for the least secure setpoints. If the PowerWizard is currently at the required level of protection when viewing a parameter, the padlock will not appear.

If a parameter is displayed with a padlock but no security level identification number next to it, the parameter cannot be changed from the PowerWizard display and the service tool must be used. Level 1 and Level 2 passwords are disabled when shipped from the factory. Level 1 and Level 2 passwords are user level passwords and can be used if desired.

The PowerWizard 2.0 also has a SCADA password, which can be used to secure remote communications.

To view the security menu:

MAIN MENU > CONFIGURE > SECURITY

At the top of the security menu the current security level is displayed. Within the security menu are the following options:

DROP TO MINIMUM LEVEL – used to return the current security level to the lowest level set-up. Highlight and press enter to drop to minimum security level. If no Level 1 or 2 passwords are set-up the minimum level will be 2. If a Level 2 password is set-up, the minimum level will be 1 and if a Level 1 password is set up the minimum level will be 0.

ENTER LEVEL 1 OR 2 – used to enter Level 1 or 2 passwords. Highlight and press enter to proceed to the password entry screen. Passwords can be entered using the cursor keys. In PowerWizard, Level 1 and 2 passwords must be different. An entered password is compared against the stored Level 1 and 2 passwords, if the password is correct the PowerWizard will go to the corresponding security level.

ENTER LEVEL 3 – used to obtain Level 3 access. The Level 3 security password is reserved for critical setpoints that should only be changed by a skilled operative. The Level 3 password is a prompt and response password.

Highlight and press enter to proceed to the phone in prompt display. The Level 3 password can be obtained by contacting the genset manufacturer and providing the 16 digit phone in prompt. The manufacturer will then provide the relevant response. To enter the 16 digit response press enter again. Passwords can be entered using the cursor keys. The Level 3 password will expire 10 minutes after the last key pressed.

The Level 3 password can be obtained from the After Sales Helpdesk. Refer to the contacts list at the back of the manual.

CHANGING LEVEL 1 PASSWORD – used to set-up, change or disable a Level 1 password. In order to use this feature the control must be at current security Level 1 or higher. Highlight and press Enter to proceed to the password entry screen. To set up or change the password, enter the new password using the cursor keys. Passwords may be 16 digits long. To disable the Level 1 security password, set the password to "0". Press the Enter key to save.

CHANGING LEVEL 2 PASSWORD – used to set up, change or disable a Level 2 password. In order to use this feature the control must be at current security Level 2 or higher. Highlight and press enter to proceed to the password entry screen. To set up or change the password, enter the new password using the cursor keys. Passwords may be 16 digits long. To disable the Level 2 security password, set the password to "0". Press the Enter key to save.

CHANGING SCADA PASSWORD (PowerWizard 2.0 only) – used to set-up, change or disable a SCADA password. Highlight and press enter to proceed to the password entry screen. To set-up or change the password, enter the new password using the cursor keys. Passwords may be 16 digits long. To disable the SCADA security password, set the password to "0". Press the Enter key to save.

4.4 Real Time Clock Programming (PowerWizard 2.0)

The real time clock provides information for the time and date of an automatic time based start/stop control. It also provides a mechanism for time stamps in the event log. The real time clock is not calibrated and is for information only. The date and time are set by the user.

1. In order to set the time or date format:
MAIN MENU > CONFIGURE > TIME/DATE
2. To set the time, highlight the time then press the Enter key twice.
3. Use the cursor keys to set the time and press the Enter key to save. Press the Enter key to return.
4. To set the date, highlight the date then press the Enter key twice.
5. Use the cursor keys to set the date and press the Enter key to save. Press the Enter key to return.
6. To set the date format, highlight either the 'FORMAT DD/MM/YY' or 'FORMAT MM/DD/YY' and press the Enter key.
7. Use the cursor keys to select the required date format and press the Enter key to save.

4.5 Fuel Priming – Engines with Electronic Fuel Lift Pump

Certain engines fitted with an electronic fuel pump do not have a manual priming feature on the engine. In these circumstances the PowerWizard can be used to energise the fuel lift pump in order to prime the engine.

1. In order to prime the genset:
MAIN MENU > CONTROL > ENGINE FUEL PRIMING
2. To prime the set press the right cursor key, this will initiate a 180 second priming cycle.
3. To exit the priming cycle press the left cursor key.

Note – The generating set may only be primed when the set is stopped and there are no active or present shutdown conditions.

4.6 Programmable Cycle Timer (PCT) (PowerWizard 2.0)

To utilise the Programmable Cycle Timer feature, one may require a software upgrade. To confirm if a software upgrade is required, please contact the after sales help desk. Refer to the contact list at the back of this manual.

The Programmable Cycle Timer (PCT) feature allows the operator to program times when two independent tasks, called PCT outputs, will be activated or deactivated automatically during the week.

This is useful for cases where two or more generating sets are required to automatically share the duty of supplying a load throughout the week. Using programmable cycle timer, each generating set can be programmed to start and stop at pre-set times. The cooperation of a transfer switch is required to ensure that the generating sets are not stopped on load.

Programmable Cycle Timer can also be used in cases where a standby set does not have access to a utility supply to power a trickle charger. Programmable Cycle Timer can be used to run the set for an hour a week to keep the battery well charged.

The PCT feature consists of seven independent timers. Each timer has the following setpoints (setpoints shown are for PCT #1):

- Programmable Cycle Timer #1 Activation Day of the Week. This permits independent selection of each day (Sun > Sat) that the timer will activate
- Programmable Cycle Timer #1 Activation Start Time. The time of day (in hours and minutes) that the timer will activate
- Programmable Cycle Timer #1 Active Time. The duration (in hours and minutes) for which the timer will be active (up to 24 hours)
- Programmable Cycle Timer #1 Output #1 Activation Configuration. Determines the first output will be activated when this timer is active
- Programmable Cycle Timer #1 Output #2 Activation Configuration. Determines the second output will be activated when this timer is active.

Output #1 is used to run the genset. Output #2 can be used to drive relay output 8. Please see the Relay Outputs section for more information.

4.6.1 PCT Status

The status of the PCTs can be found within the menu structure at:
MAIN MENU > CONTROL > PROG CYCLE TIMER:

- The top row displays the seven PCTs (#1 to #7).
- The middle row is for OUTPUT #1.
- The bottom row is for OUTPUT #2.

4.6.2 Controlling The Outputs

The status of each output is indicated by a 1, 0 or - in the "PROG CYCLE" column under #1 to #7"

- "1" The PCT is configured and currently driving the output.
- "0" The PCT is configured but not currently driving the output.
- "-" The PCT has not been configured:

PROG CYCLE	:	1	2	3	4	5	6	7
OUTPUT 1	:	1	0	-	-	-	-	-
OUTPUT 2	:	-	0	0	-	-	-	-

4.6.3 PCT Example:

- PCT #1 is ACTIVE, "1" – Output #1. Output #2 is not enabled (the "-" in the bottom row)
- PCT #2 is NOT active, "0" – however, when it is active, it activates both Output #1 and Output #2. The two zeros show that this PCT will enable both outputs when it becomes active
- PCT #3 is not active. When it becomes active it will only activate Output #2
- PCT #4, PCT #5, PCT #6 and PCT #7 are disabled. They will not activate Output #1 or Output #2.

5 Installation

5.1 Power Requirements

The PowerWizard series of generating set controls require a nominal voltage of 12 Vdc or 24 Vdc. If batteries are used for operating power, a charging source such as an alternator or battery charger is necessary to maintain a stable supply voltage. Under steady state operation, the PowerWizard controllers on 12V sets have approximately an 850 m Amp current draw (not including any relay loads).

This current drain can be reduced by approximately a factor of 7 by using the Reduced Power Mode option (RPM). However it is recommended that generating sets at rest or storage for prolonged periods should have either the battery charger or isolator switch option fitted.

When connecting the PowerWizard generating set control to the DC power source, make sure that there is only one common connection to the negative potential of the power source. Make extra effort to avoid any ground loops in the DC electrical system. A single point common ground for electronics is recommended at the negative battery terminal or Power Distribution Box. Each electronics sub-system and major engine sub-system should have its own DC network so that they do not interfere with each other. An example is shown in Figure 7.

As shown in the figure all electronics are electrically isolated from higher current loads, such as the starter motor. All electronics have a common Power Bus and Single Point Reference. The chassis ground is a common Power and Transient Ground.

The electronics, such as sensors and control modules, have isolated power source paths. High current loads such as starters and solenoids can cause interference and possibly damage to low current loads, such as controllers and sensors. Extra effort must be made to keep the high current and low current loads electrically separated. The two types of loads may share common (+)Battery and (-)Battery connections, but they should not be electrically connected. This strategy ensures maximum isolation between high and low current loads.

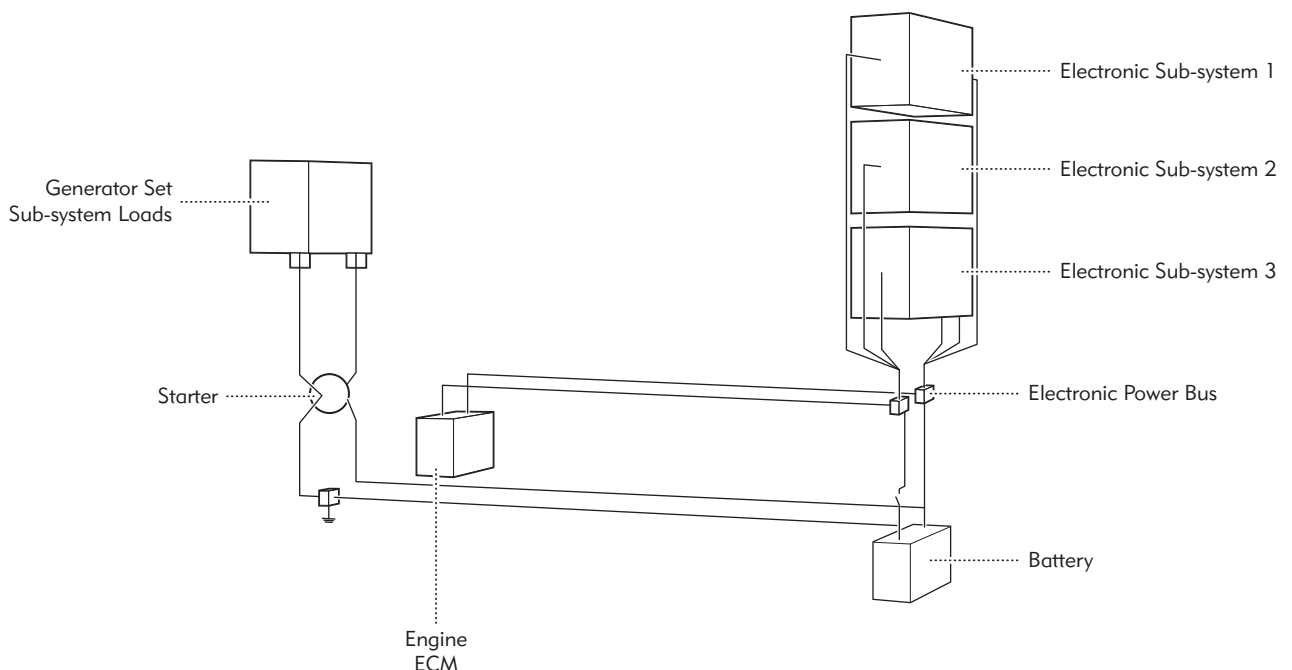


Figure 7: Generating Set Network Isolation

The battery disconnect switch is located on the negative leg of the battery supply. If a battery charger is to be used, it should be connected on the battery side of the disconnect switch, so as not to power the electronics. Most battery chargers are not to be used as power supplies. Proper battery charger operation requires that the actual battery load is present.

5.2 Location Considerations

When selecting a location for mounting the PowerWizard generating set control, consider the following:

- Protection from high-voltage and high-current devices
- Protection from devices that may produce electromagnetic interference
- Protection from excessive vibration. The controls are designed to withstand normal generating set vibrations. The controls should not be mounted directly to the engine block
- Protection from direct exposure to water. Once installed, the PowerWizard 1.0 and 2.0 controls are sealed to a level of IP Level 22 for resistance to moisture
- The continuous operating range of the PowerWizard generating set controls is -20 to $+70^{\circ}\text{C}$ ambient.

5.3 Electrical Connections

The PowerWizard control has one 70-pin connector on the back of the control. Not all 70 pins are used. The following diagrams show what pins are used and what each pin should be connected to for each version of the control.

Figures 8 and 9 are shown with all possible connections used. For Electronic Engines (EUI), the passive analogue inputs number 1 and 2 will not be used. These are for oil pressure and coolant temperature respectively. On EUI engines, those sensors will be wired to the engine ECM and the PowerWizard will get that information from the engine ECM via the Primary J1939 Data Link.

The method used for the analogue inputs is 1-wire sensors as shown in the diagram.

The discrete inputs are shown connected through normally open contacts to battery negative. These inputs can also be connected through normally closed contacts to battery negative. In order to do this the active state of the input will need to be set to active high.

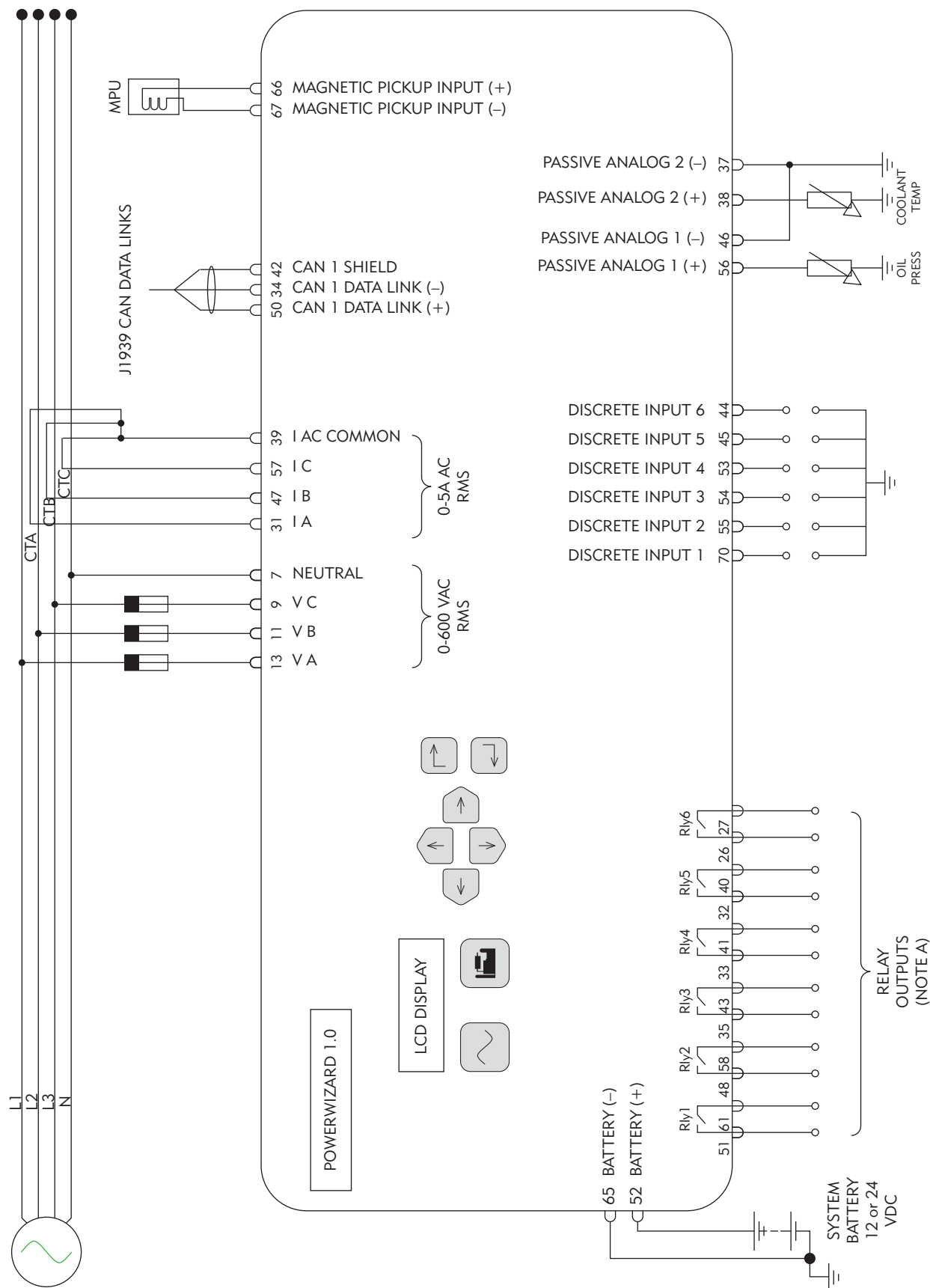


Figure 8: PowerWizard 1.0 Control Electrical Connections

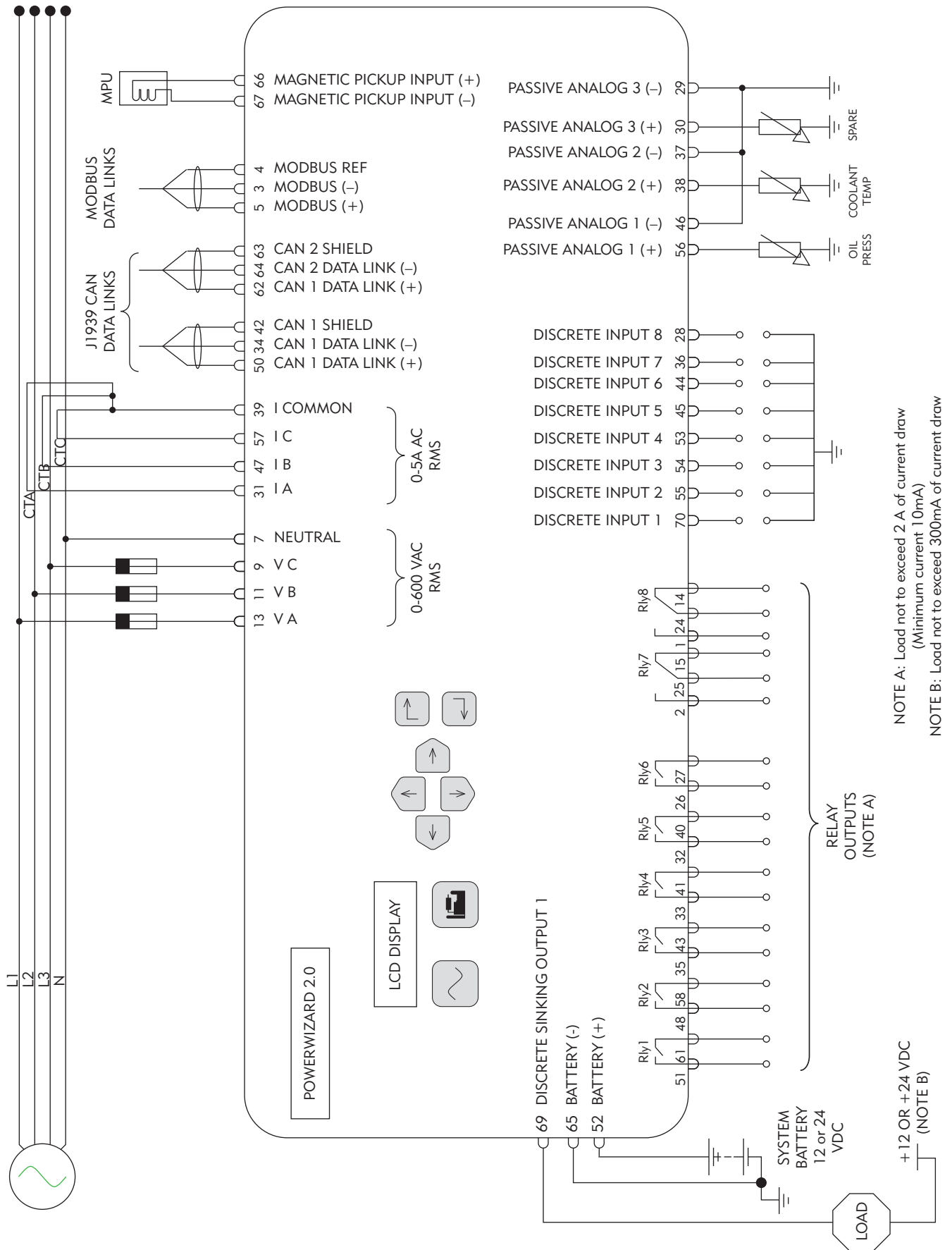


Figure 9: PowerWizard 2.0 Control Electrical Connections

5.4 Transformer Connections

The PowerWizard can monitor generating set output voltages in the range of 80 VAC to 600 VAC. In order to monitor voltages greater than 600 Volts, external potential transformers must be used.

Note – The PowerWizard must be programmed for the correct winding ratios when connecting external potential transformers.

Note – The wye configuration of external potential transformers is preferred for 4-wire wye generating sets because of the greater accuracy when loads are unbalanced. With the open delta configuration, some power parameters cannot be determined. These parameters are real power phase A, B, C and power factor phase A, B, C. For maximum accuracy, the open delta configuration of external potential transformers should be used only for 3-wire delta generating sets.

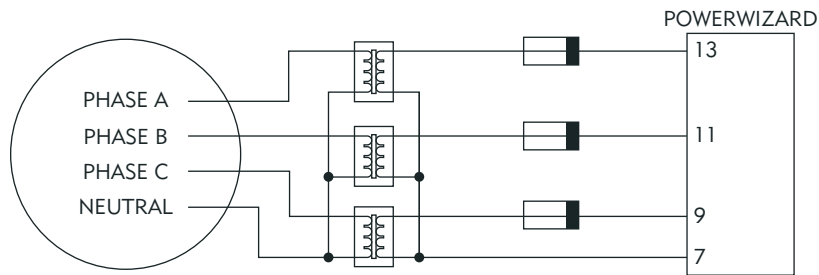


Figure 10: Wye Configuration of External Potential Transformers (PT) on the 4-Wire Wye Connected Generating Set

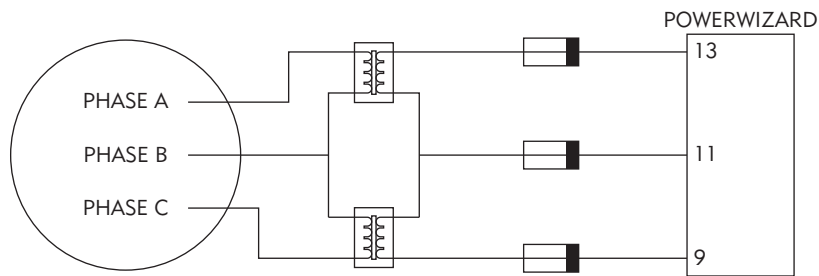


Figure 11: Open Delta Configuration of External Potential Transformers (PT) on the 3-Wire Delta Connected Generating Set

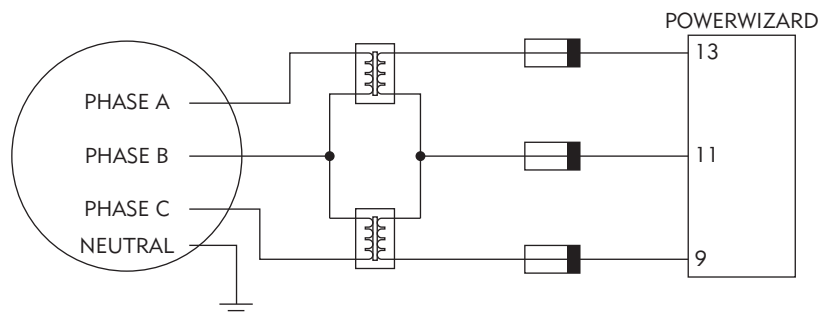


Figure 12: Open Delta Configuration of External Potential Transformers (PT) on the 4-Wire Wye Connected Generating Set

5.5 Data Links

The PowerWizard supports up to 3 different data links:

- One Primary J1939 Data Link
- One Accessory J1939 Data Link (PowerWizard 2.0 only)
- One System Control and Data Acquisition (SCADA) Data Link (PowerWizard 2.0 only).

For information on the SCADA system refer to the following manuals:

- PowerWizard Monitoring Software (available with MCM7 and MCM8 options)
- PowerWizard Modbus Applications Guide (provided by the after sales help desk, refer to the contacts list at the back of this manual.

5.5.1 Primary J1939 Data Link

The Primary J1939 Data Link is supported by all of the PowerWizard controls.

The Primary J1939 Data Link is used for local communication among modules associated with a single generating set. The PowerWizard can interface with both Electronic Engines (EUI) and Mechanical Engines (MUI). In MUI engines, the engine sensors are wired directly to the PowerWizard. The Primary J1939 Data Link utilizes the Society of Automotive Engineers (SAE) J1939 protocol and requires hardware compliant to the high-speed Controller Area Network (CAN) 2.0B protocol defined in the International Standards Organization (ISO) 11898-2 document, running at 250k bits per second.

The Primary J1939 Data Link supports appropriate SAE J1939 Broadcast Parameter Group Numbers (PGN) and Suspect Parameter Numbers (SPN) for engine and generating set data and diagnostics.

Wiring

The Primary J1939 communication wires are brought out of the PowerWizard as part of the 70-pin AMP connector. The pins, as designated on the AMP connector, are shown in Table 2.

Pin #	Name	Description
34	CAN1 –	Differential (–) for CAN
42	CAN1 SH	Shield for CAN
50	CAN1 +	Differential (+) for CAN

Table 2: Primary J1939 Data Link on 70-pin Connector

Network Topology

The physical topology of the CAN network used in the Primary J1939 Data Link is a bus topology, consisting of a main trunk and small drops. The maximum allowable trunk length is 130 ft (40 m), and the maximum drop length is 3 ft (1 m). The CAN network requires a termination resistor on the extreme ends of the main trunk. The topology for the PowerWizard 1.0 is illustrated in Figure 13.

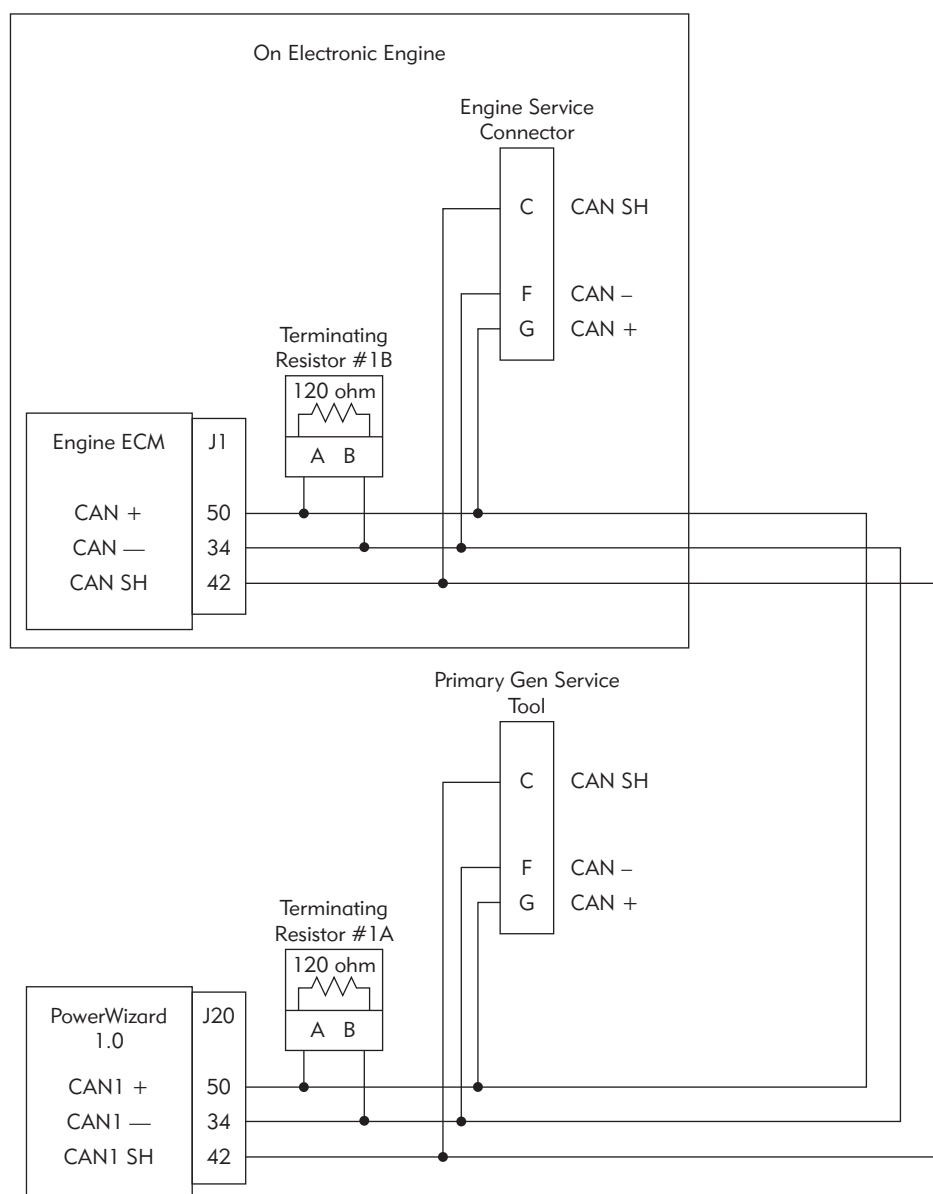


Figure 13: PowerWizard 1.0 CAN Wiring Diagram

5.5.2 Accessory J1939 Data Link

The Accessory J1939 Data Link is supported by the PowerWizard 2.0.

The Accessory Data Link is used for local or remote communication among modules associated with a single generating set. This includes up to three annunciators. The Accessory J1939 Data Link utilizes the Society of Automotive Engineers (SAE) J1939 protocol and requires hardware compliant with the high-speed Controller Area Network (CAN) 2.0B protocol defined in the International Standards Organization (ISO) 11898 2 document, running at 250k bits per second.

The Accessory J1939 Data Link supports appropriate SAE J1939 Broadcast Parameter Group Numbers (PGN) and Suspect Parameter Numbers (SPN) for engine and generating set data and diagnostics.

Wiring

The Accessory J1939 communication wires are brought out of the PowerWizard 2.0 as part of the 70-pin AMP connector. The pins, as designated on the AMP connector, are shown in Table 3:

Pin #	Name	Description
62	CAN2 +	Differential (+) for CAN
63	CAN2 SH	Shield for CAN
64	CAN2 –	Differential (–) for CAN

Table 3: Accessory J1939 Data Link on 70-pin Connector

Network Topology

The physical topology of the CAN network used in the Accessory J1939 Data Link is a bus topology, consisting of a main trunk and small drops. The maximum allowable trunk length is 800 ft (244 m), and the maximum drop length is 3 ft (1 m). The CAN network requires a termination resistor on the extreme ends of the main trunk. The topology for the PowerWizard 2.0 with some remote devices connected is illustrated in Figure 14. Note that more remote devices can be connected, as long as the proper lengths are maintained, and the termination resistor is placed at the end of the trunk.

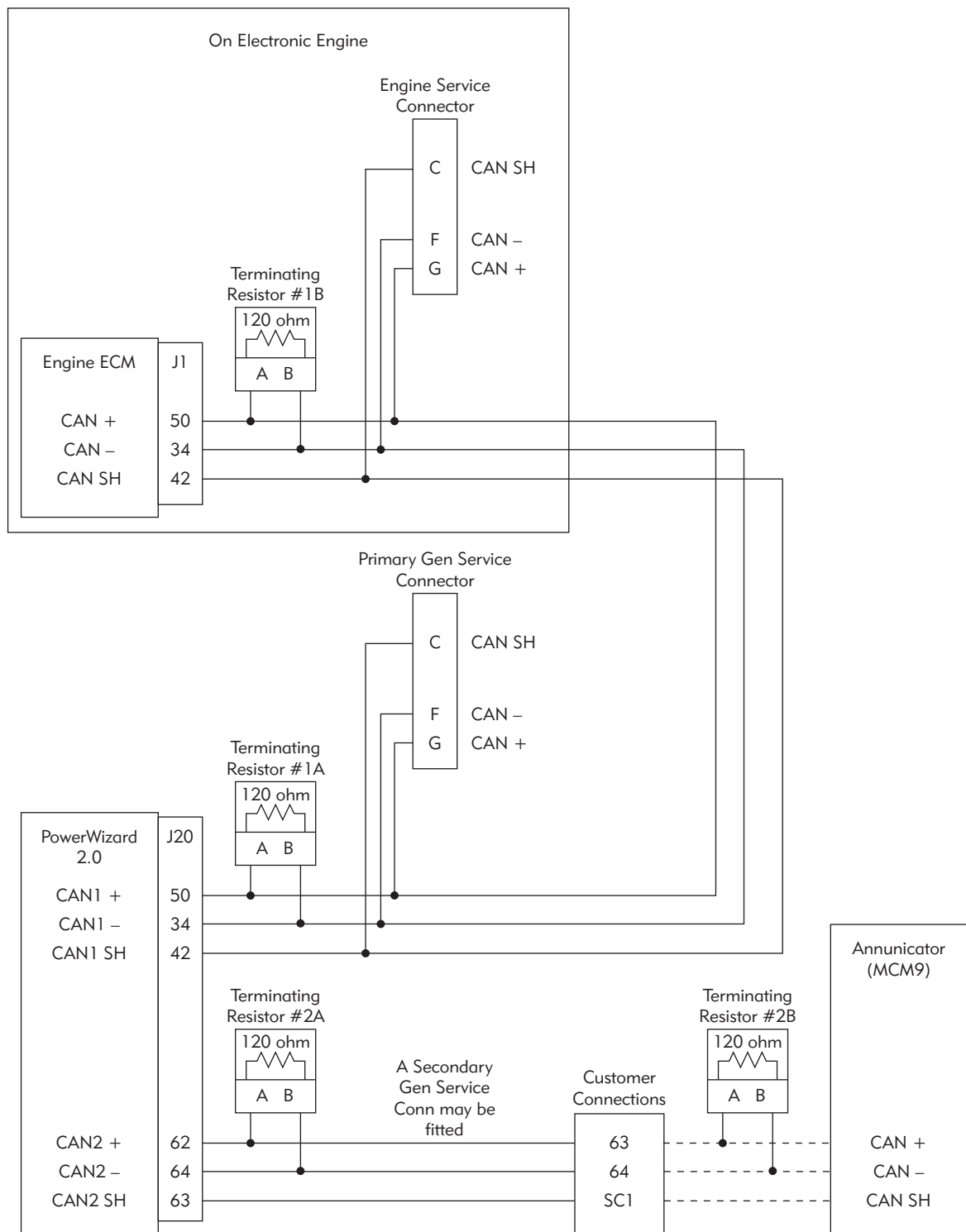


Figure 14: PowerWizard 2.0 CAN Wiring Diagram (with optional module)

5.6 Optional Modules

5.6.1 Annunciator (MCM9)

The PowerWizard Annunciator serves to display generating set system alarm conditions and status indications. The Annunciator has been designed for use on the PowerWizard J1939 Communication Network.

It is used in remote applications being mounted separate from the generating set to provide remote indication of system operating and alarm conditions.

The PowerWizard Annunciator is configurable to the standards of NFPA 99/110.



5.6.2 Annunciator Features

- Each Annunciator includes 16 LED pairs for annunciation of up to 32 system events
- An additional pair of LEDs provides status indication of the J1939 data link connection
- Can annunciate alarm conditions received from any module on the J1939 data link, including the PowerWizard and engine ECM
- Includes alarm horn with lamp test and alarm acknowledge pushbuttons
- Configurable to NFPA 99/110 requirements for remote annunciation on emergency standby generating set systems
- Provides a simple means to change the labels for the annunciation LEDs for site specific requirements
- Provides superior visibility of the LEDs in direct sunlight
- Graphic symbols are provided next to each LED pair to indicate various alarms and events
- The Annunciator can be mounted remotely (up to 800 feet) on the accessory J1939 data link
- Designed and tested to meet stringent impulse shock and operating vibration requirements.

5.6.3 Annunciator Specifications

Power Supply

Accepts 9 - 32VDC power (12 or 24VDC nominal).
Over voltage capability is 32VDC for 1 hour @ 85°C.

Packaging and Dimensions:

- -40 to 85°C operating temperature range (for ambient temperatures exceeding 85°C, the temperature scanner may deviate in accuracy an additional $\pm 1^\circ\text{C}$. Note also that 120°C is a failure point. If the ambient temperature were to exceed 120°C, the device would NOT be expected to return to proper operation)
- The ambient storage temperature range is -50°C to +120°C
- It is protected against 95% humidity non-condensing, 30°C to 60°C
- Designed to meet relevant European standards for EMI/RFI/Immunity without the use of external filtering (Third party testing is being pursued).

Dimensions:

- 286 mm (11.26 in) High
- 157 mm (6.18 in) Wide
- 141 mm (5.55 in) Deep

Communications Interface:

- Incorporates an SAE J1939 communications port
- Node address is auto configurable as per J1939-81
- Optical isolation is provided for the CAN line
- Module operates normally with loss of communication link, retaining configured setpoints in non-volatile memory
- Configuration is accomplished with customer proprietary service tools over the J1939 network.

SAE J1939 Profile:

- For J1939 compliance the module complies with the applicable portions of the following:
 - SAE J1939-21, July 1998, Data Link Layer
 - SAE J1939-71, January 2002, Application Layer
 - SAE J1939-73, May 2000, Application Layer – Diagnostic
 - SAE J1939-81, July 1997, Network Management
- Customer specific proprietary extensions are also included in the SAE J1939 profile.

Connections to Annunciator using terminals at back of Annunciator.

Connections to the Annunciator must be hard wired as shown below. (The plug is not used):

- Terminal 8: Can-H
- Terminal 9: Can-L
- Terminal 10: Shield
- Terminal 11: PWR+
- Terminal 12: PWR-

5.6.4 Annunciator LED Colour Scheme

Each pair of LEDs on the Annunciator consists of two of the following three colours: Green, Yellow and Red, which allows for custom configuration of Status, Warning and Shutdown conditions. The available colours and combinations are as follows:

Row	LED1	LED2
1	Red	Yellow
2	Red	Yellow
3	Red	Yellow
4	Red	Yellow
5	Red	Yellow
6	Red	Yellow
7	Red	Yellow
8	Red	Yellow
9	Red	Yellow
10	Red	Yellow
11	Red	Yellow
12	Red	Yellow
13	Green	Yellow
14	Green	Yellow
15	Red	Green
16	Red	Green

Table 4: Available colours and combinations of LED pairs

5.6.5 Annunciator Configuration

The Annunciator is field programmable using the Electronic Service Tool. It is also flash programmable to update software using the service tool.

Information on purchasing and licensing the Electronic Service Tool (EST) can be found on the Dealer website or by contacting the EST Software License and Support Coordinator. Refer to the contacts list at the back of the manual.

The service tool software must be installed on a Windows PC. The communication adapter must be connected between the PC and the J1939 data link on which the target Annunciator is connected. (The service tool may be connected to the PowerWizard Accessory Data Link service connector).

When connecting to the Annunciator, the user will first see the Module Summary screen shown in Figure 15.

This screen shows module information such as Serial Number, Part Number, Software Group Number and Software Release Date.

The Module Description will also indicate which of the 3 possible Annunciators the service tool is connected with. In this example, the service tool is connected to "Alarm Module #2". This number is called the "ECU Instance" and it is programmable. To program the Annunciator to a different ECU Instance, enter the Configuration Screen by selecting the Service → Configuration menu item, as shown in Figure 15.

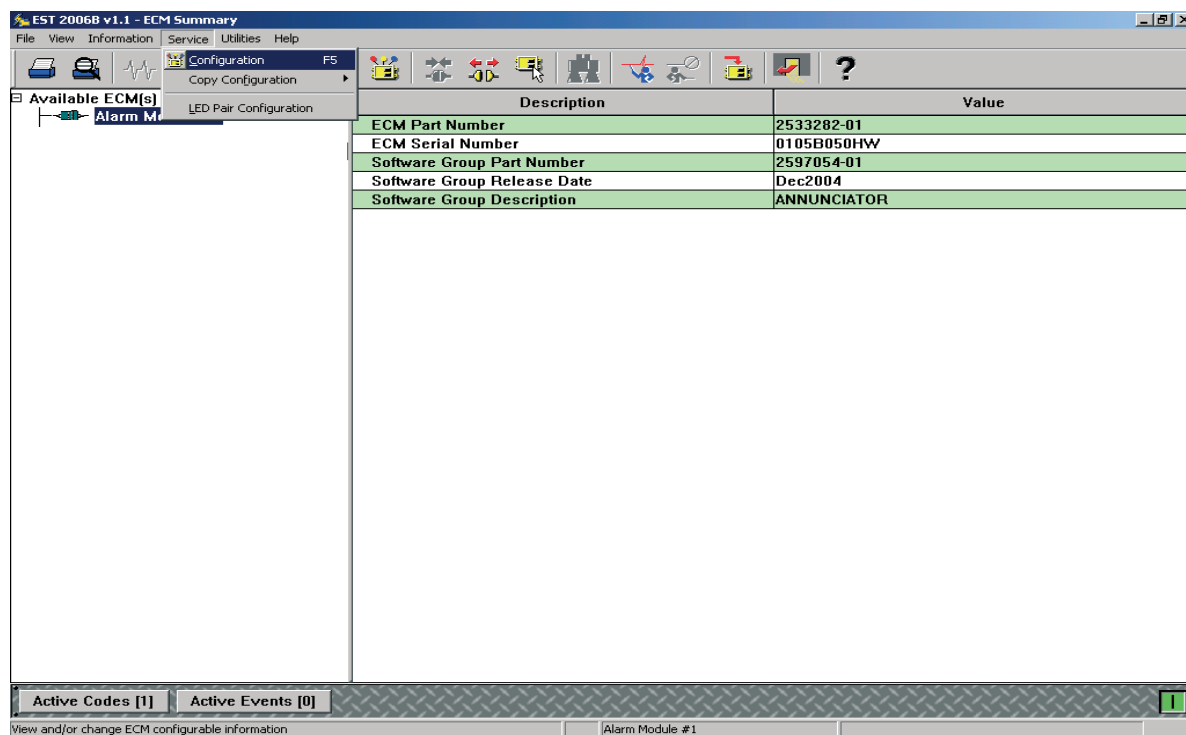


Figure 15: Service Tool Annunciator Summary Screen

The service tool configuration screen, shown in Figure 16 identifies the serial number of the Annunciator that is connected. This is important in matching the desired ECU Instance to the actual hardware. In this example, the Annunciator will be programmed to ECU Instance #1. To do this, select the ECU Instance row, and double click on the current ECU Instance. A dialog box will open, allowing entry of the new ECU Instance. Type a numeric value, 1, 2 or 3 and click OK.

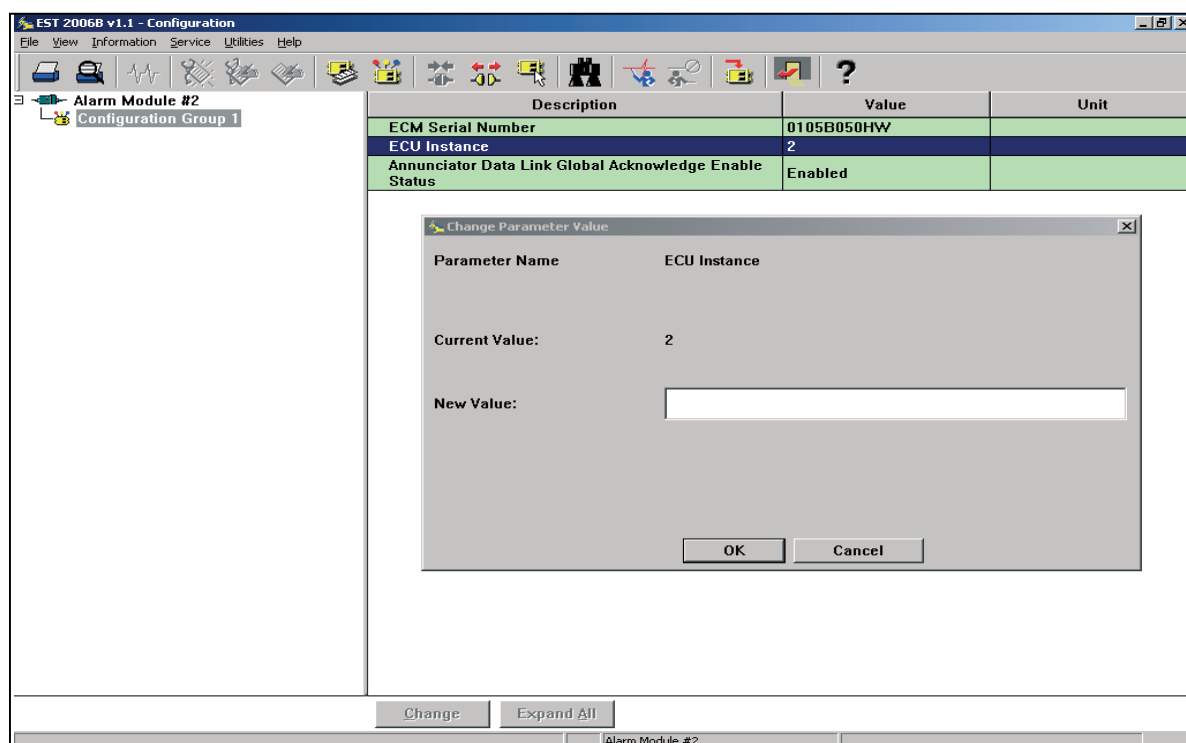


Figure 16: Service Tool Annunciator Configuration Screen

When you click OK, the service tool software will automatically restart and reconnect to the data link. Upon reconnecting, the summary screen will now show the new module name, based on the ECU Instance. In this example, the new module name will be "Alarm Module #1".

5.6.6 Annunciator Global Acknowledge

The Annunciator can be configured to both initiate and respond to an Event Acknowledge message over the J1939 data link. If this setpoint is enabled, the Annunciator events may be acknowledged remotely by pressing the Alarm Acknowledge button on a PowerWizard or by pressing the Acknowledge button on another Annunciator that is on the same data link. The default setting for this setpoint is DISABLED, but it can be enabled from the Service → Configuration menu item.

5.6.7 Configuring Annunciator LED Behaviour

To configure the behaviour of the LED pairs, enter the LED Pair Configuration screen by selecting the Service → LED Pair Configuration menu item.

An example of the LED Pair Configuration screen is shown in Figure 17.

LED Pair #	Trigger Condition	Severity Level	Suspect Parameter Number	Failure Mode Identifier
LED Pair # 1	Specific Event	Condition Met, Red On, Other Not Used, Horn On	Emergency Stop Shutdown [970]	
LED Pair # 2	Specific Event	Condition Met, Red On, Other Not Used, Horn On	Engine Failure to Start Shutdown [1664]	
LED Pair # 3	Specific Event	High Warning or High Shutdown	Engine Coolant Temperature [110]	
LED Pair # 4	Specific Event	Low Warning or Low Shutdown	Engine Oil Pressure [100]	
LED Pair # 5	Specific Event	High Warning or High Shutdown	Engine Speed [190]	
LED Pair # 6	Specific Event	Low Warning or Low Shutdown	Engine Coolant Level [111]	
LED Pair # 7	Specific Event	Low Warning	Battery Voltage [168]	
LED Pair # 8	Specific Event	Low Warning	Unknown SPN [167]	
LED Pair # 9	Specific Event	Condition Met, Amber On, Other Not Used, Horn On	Unknown SPN [4007]	
LED Pair # 10	Specific Event	Low Warning	Fuel Level (Local Tank) [96]	
LED Pair # 11	Specific Event	High or Low Warning, High or Low Shutdown	Generator AC Voltage [2440]	
LED Pair # 12	Specific Event	High or Low Warning, High or Low Shutdown	Generator Frequency [2436]	
LED Pair # 13	Specific Event	Condition Met, Green On, Other Not Used	Unknown SPN [4002]	
LED Pair # 14	Specific Event	Condition Met, Green On, Other Not Used	Unknown SPN [4001]	
LED Pair # 15	Specific Event	Condition Met, Red On, Other Not Used	Unknown SPN [4017]	
LED Pair # 16	Disabled	High or Low Shutdown		

Submit

Alarm Module #1

Figure 17: Service Tool Annunciator LED Configuration Screen

Each LED pair has four parameters required to configure it. It is best to configure the four columns in order from left to right; first Trigger Condition, next Severity Level, then Suspect Parameter Number, and finally Failure Mode Identifier (if required). The reason for the order is because the value set in one column affects the choices available in the subsequent columns. If a field is greyed out, it is not available due to conditions set in previous columns.

Trigger Condition: There are three possible selections for Trigger Condition: Specific Event, General Event and Disabled.

“Specific Event” is used to assign an LED pair to a specific data link parameter such as Oil Pressure, Engine Speed, Coolant Temperature, etc.

“General Event” is used to assign an LED pair as a general alarm or shutdown indicator. When configured as General Event, the LED will not be assigned to a particular parameter. It will respond to any event, regardless of the Suspect Parameter Number. For this reason, when General Event is selected, the Suspect Parameter Number cannot be changed.

“Disabled” is used to disable the LED pair. The remaining three parameters will be greyed out when Disabled is selected.

Severity Level: Severity Level defines how the LED pair will react to various levels of event conditions. Selections that begin with “Condition Met” will respond to J1939 Event messages for FMI 31 “Condition Present”. For example, LED pair #1 is configured for “Condition Met, Red On, Other Not Used, Horn On” with SPN 970 (Emergency Stop Active). This means that when the Annunciator receives a J1939 message indicating Emergency Stop with FMI 31, the Red LED will turn on and the Horn will also turn on. “Other Not Used” indicates that the other LED colour in the pair is never used. “Green Off” indicates that the green LED (for example) lights when the condition chosen for this LED pair is NOT active.

The Severity Level selections imply a J1939 Failure Mode Identifier (FMI) code. “Condition Met” is equivalent to FMI 31. High Warning can be FMI 15 or FMI 16. Low Warning can be FMI 17 or FMI 18. High Shutdown is equivalent to FMI 0. Low Shutdown is equivalent to FMI 1. The FMI column is greyed out for any of these selections. The only option that will allow an FMI to be configured is “Specific Diagnostic Code”.

Suspect Parameter Number: The SPN column is used to type in the Suspect Parameter Number for the parameter assigned to the LED pair. Refer to the SAE J1939 literature for a complete list of supported SPNs.

Note – Remember to click the Submit button at the bottom of the LED Pair Configuration screen after making the selections for each LED pair. Only then will the new settings take effect.

5.6.8 PowerWizard Support of the Annunciator

Note – CAN1 indicates the connection for the PowerWizard Primary J1939 data link, and CAN2 indicates the connection for the PowerWizard Accessory J1939 data link. The PowerWizard 1.0 only supports the Primary J1939 data link. For more information on the Primary and Accessory data links, refer to the Data Links section.

PowerWizard 1.0: Supports one Annunciator module using CAN1.

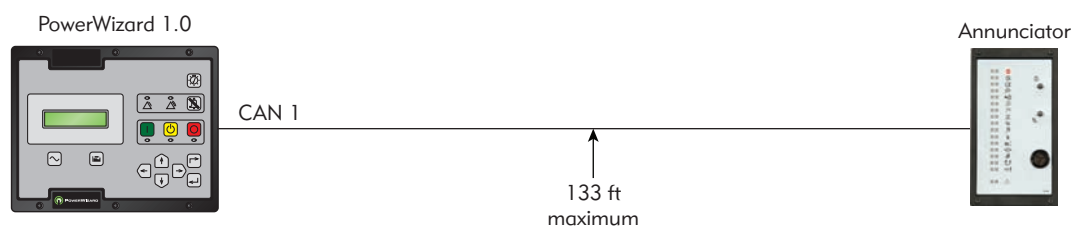


Figure 18: PowerWizard 1.0 and Annunciator Connection

PowerWizard 2.0: Supports one Annunciator module using CAN1 and up to three Annunciators using CAN2.

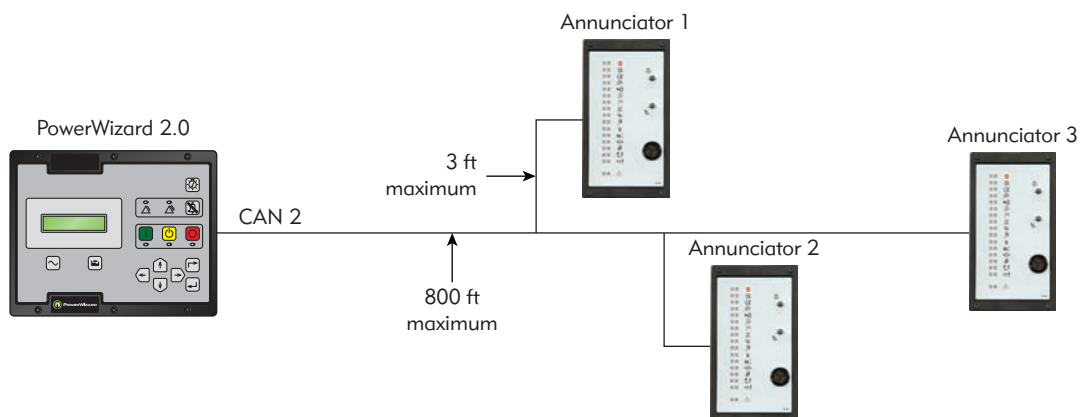


Figure 19: Example illustration of Accessory Data Link Modules and Wire Lengths

6 Setpoint Programming

The engine/generating set setpoints affect the proper operation and serviceability of the engine, and the accuracy of information shown on the display screen. The setpoints are programmed in the PowerWizard at the factory.

The setpoints may require changing when the PowerWizard is moved from one engine to another. The setpoints may also require changing in order to satisfy the requirements of the installation. The setpoints that are stored in the PowerWizard must match the specified setpoints of the particular generating set. For a list of all available setpoints see Appendix B.

6.1 Digital Input Programming (Spare Fault Input)

Digital inputs are used to inform the PowerWizard of warnings or shutdowns that exist external to the module. Digital inputs are used for Emergency stop and remote start amongst various other parameters. There are two parts to programming the Digital Inputs. The first part involves programming the Active State of the Digital Input (Active High or Active Low). The second part involves programming the Event Input Functions.

Digital Input #1 is dedicated to Emergency Stop. Digital Input #2 is dedicated to Remote Initiate (Remote Start). The six remaining Digital Inputs on the PowerWizard 2.0 (four remaining Digital Inputs on PowerWizard 1.0) can be programmed for various other applications.

6.1.1 Programming the Active State of the Digital Input

The digital inputs of the PowerWizard are tied to an internal pull-up resistor. Therefore, if there is no connection to a digital input, the digital input will sit at a logical high. A ground or -batt input should be used to trigger a Digital Input. If an Active High configuration is desired, the ground or -batt input should be wired through a normally closed switch. If an Active Low configuration is desired, the ground or -batt input should be wired through a normally open switch.

1. To program the Active State of the Digital Input, go through the following menu options:
MAIN MENU > CONFIGURE > SETPOINTS > I/O > DIGITAL INPUTS.
2. Select the Digital Input that you want to program. Press the Enter key.
3. Press the Enter key again. The current configuration (either 'HIGH' or 'LOW') will be highlighted.
4. Use the Scroll Up key and the Scroll Down key in order to change the current configuration to the desired setting (either 'HIGH' or 'LOW').
5. Press the Enter key to save the setting.

6.1.2 Programming the Event Input Functions

Events in PowerWizard are defined as Suspect Parameter Number (SPN) and Failure Mode Identifier (FMI) combination. Essentially the SPN is the system or component at fault and the FMI describes how it is at fault. A list of all supported SPN/FMI combinations can be found in Appendix A.

1. To program the Active State of the Digital Input, go through the following menu options:
MAIN MENU > CONFIGURE > SETPOINTS > EVENTS > EVENT INPUT FUNCTIONS
2. Select the event input function to be programmed. The digital inputs are linked to the Event input functions as below.

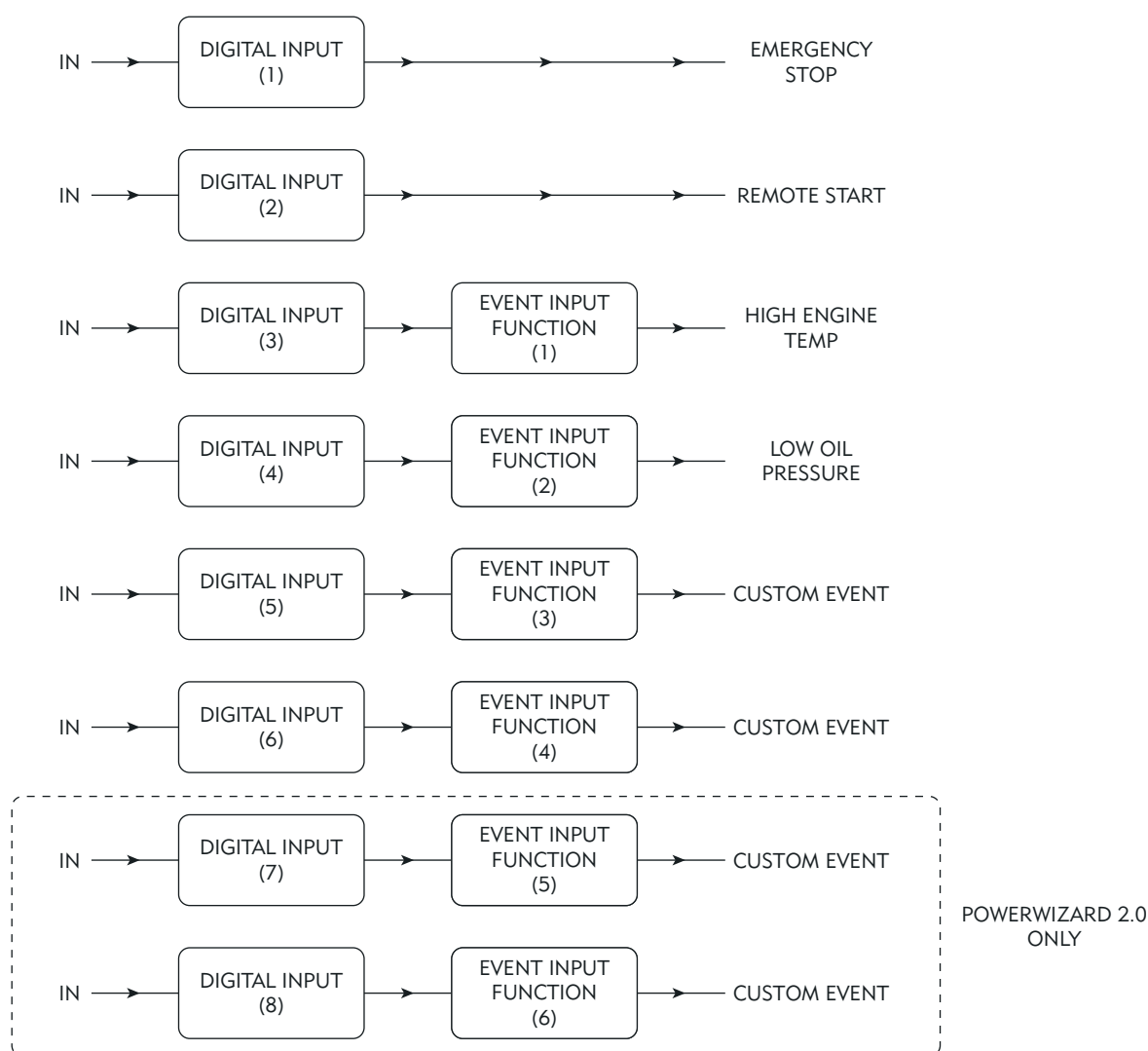


Figure 20: Digital inputs are linked to the Event Input Functions

Note – As digital inputs #1 and #2 are dedicated, Event Input Function #1 corresponds to Digital Input #3, Event Input Function #2 corresponds to Digital Input #4, etc.

3. Once in the Input Function menu, the first setting is the Active State. The Active State should always be set to Active High.
4. The next setting is the Time Delay. While on the Time Delay Setting, press Enter and use the arrow keys to enter the desired value.

5. The next setting is the Suspect Parameter Number (SPN). Press the Enter key in order to choose the SPN. In order to change event input functions 1 and 2 the service tool or Level 3 password is required. Below is a list of available SPNs.

Pressures:

- Air Filter Differential Pressure
- Engine Oil Pressure
- Fire Extinguisher Pressure
- Fuel Filter Differential Pressure
- Oil Filter Differential Pressure
- Starting Air Pressure

Temperatures:

- Ambient Air Temperature
- Engine Coolant Temperature
- Engine Oil Temperature
- Exhaust Temperature
- Rear Bearing Temperature
- Right Exhaust Temperature
- Left Exhaust Temperature

Levels:

- Engine Coolant Level
- Engine Oil Level
- Fuel Level
- External Tank Fuel Level

Others:

- Air Damper Closed
- ATS in Normal Position
- ATS in Emergency Position
- Battery Charger Failure
- Generating set Breaker Closed
- Utility Breaker Closed
- Fuel Leak Detected
- Custom Event

6. After the SPN is chosen, the Failure Mode Identifier (FMI) is the next setting. The following FMIs are available:

- **High Warning** (example: High Temperature Warning)
- **Low Warning** (example: Low Temperature Warning)
- **High Shutdown** (example: High Temperature Shutdown)
- **Low Shutdown** (example: Low Temperature Shutdown)
- **Status** (example: Fuel Tank Leak)

Status is used when an event is not essentially high or low but simply exists. An example of this is the Remote Start command.

6.2 Relay Output Programming (Spare Fault Output)

There are two parts to programming the Relay Outputs. The first part involves programming the Active State of the Relay Output (Active High or Active Low). The second part involves programming the Digital Selectors. Relay Output #1 is dedicated to controlling the starter motor. Relay Output #2 is dedicated to fuel enable.

The six remaining Relay Outputs on the PowerWizard 2.0 (four remaining Relay Outputs on PowerWizard 1.0) can be programmed for various other applications.

The PowerWizard 1.0 will have six relays: All six will be type A. The PowerWizard 2.0 will have eight relays: Six will be type A and two will be type C. Type A is defined as one normally-open contact plus common. Type C is defined as two contacts, normally-open and normally-closed plus common:

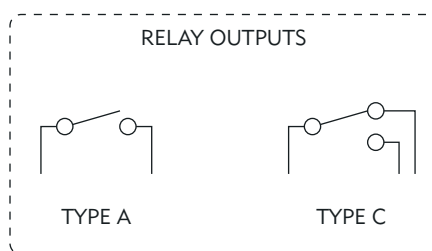


Figure 21: Relay Outputs

Each relay is capable of handling 2A @ 30 VDC. The relay contacts are not protected against shorts to battery or ground.

Note – As relay outputs #1 and #2 are dedicated, Digital selector #1 corresponds to relay output #3, Digital selector #2 corresponds to relay output #4, etc.

6.2.1 Relay Outputs

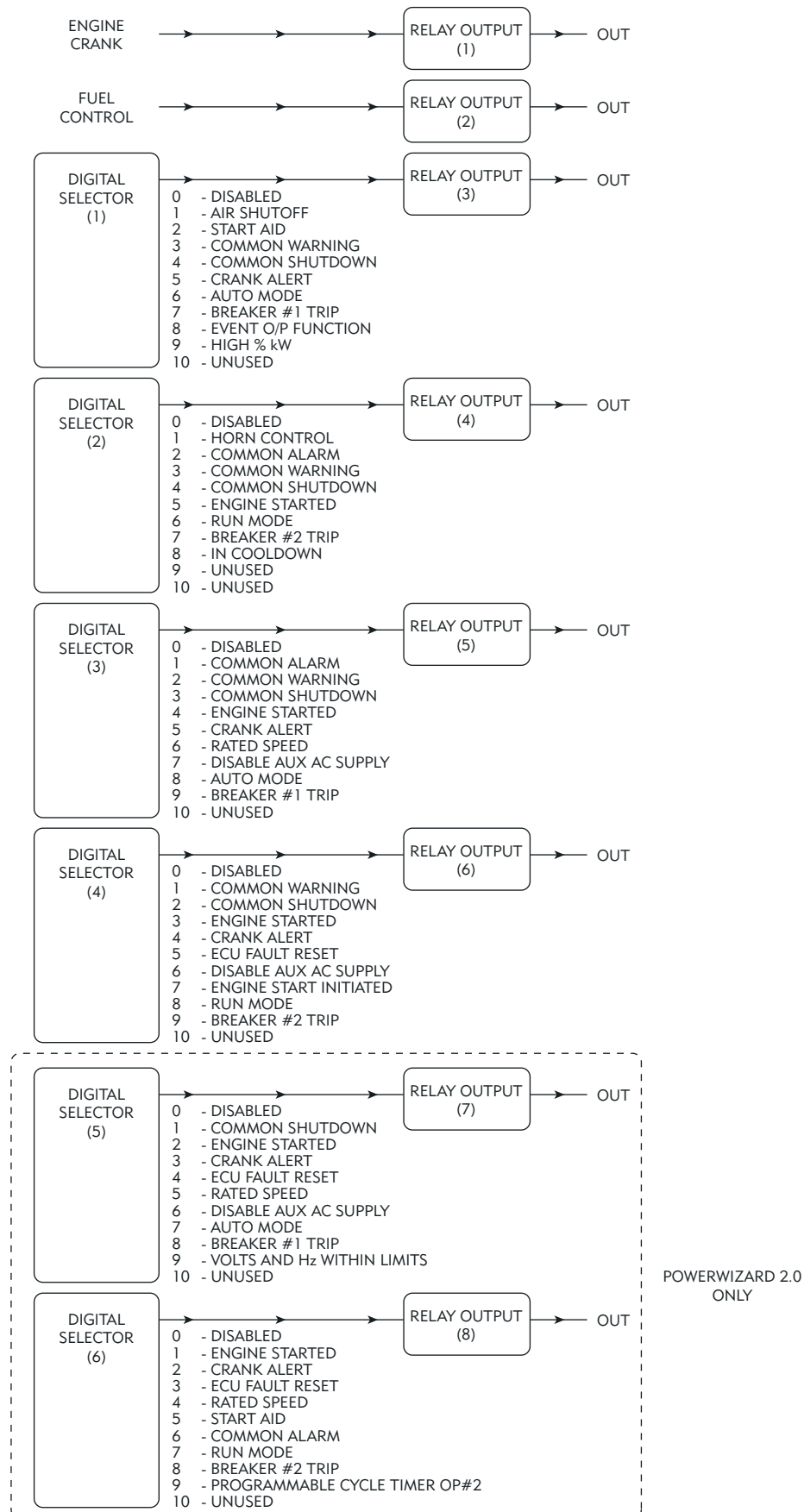


Figure 22: Digital Selectors and Relay Outputs

6.2.2 Programming the Active State of the Relay Output

The relays are internally controlled by PowerWizard. The active state is programmable but should always be set to active high. This operation can only be performed using the approved service tool.

6.2.3 Digital Selectors

The digital selectors determine what conditions cause the Relay Outputs 3 to 8 to become active. Different conditions are available on the different digital selectors.

6.2.4 Programming the Digital Selectors

1. In order to program the Digital Selectors, go through the following menu options:
MAIN MENU > CONFIGURE > OTHER > DIGITAL SELECTORS.

2. Select the Digital Selector that you want to program. Press the Enter key.

Note - As relay outputs #1 and #2 are dedicated, Digital selector #1 corresponds to relay output #3, Digital selector #2 corresponds to relay output #4, etc.

3. Press the Enter key again. The current configuration will be highlighted.

4. Use the Scroll Up key and the Scroll Down key in order to change to the value corresponding to the desired condition. (For a list of the digital selector conditions and how these correspond to values see Figure 22).

5. Press the Enter key to save the setting.

6.2.5 Event Output Function Configuration

In addition to the standard list of predefined relay output triggers there is one event output function, connected to digital selector #1. This can be used to trigger a relay output on a specific event. Firstly Digital Selector 1 should be set to event output function.

1. To program the Event Output Function, go through the following menu options:
MAIN MENU > CONFIGURE > SETPOINTS > EVENTS > EVENT OUTPUT FUNCTIONS

2. Select event output function 1. Press the Enter key.

3. Press the Enter key again to select the required trigger condition for the Event output function 1. Press the Enter key to save.

4. Scroll down to highlight the SPN parameter; press Enter to select.

5. Select the appropriate group of SPNs and press the Enter key. Below is a list of available SPNs.

Pressures:

- Air Filter Differential Pressure
- Engine Oil Pressure
- Fire Extinguisher Pressure
- Fuel Filter Differential Pressure
- Oil Filter Differential Pressure
- Starting Air Pressure
- Gas Pressure

Temperatures:

- Ambient Air Temperature
- Engine Coolant Temperature
- Engine Oil Temperature
- Exhaust Temperature
- Right Exhaust Temperature
- Left Exhaust Temperature
- Gen Rear Bearing Temperature
- Gen winding #1 Temp
- Gen winding #2 Temp
- Gen winding #3 Temp

Levels:

- Engine Coolant Level
- Engine Oil Level
- Fuel Level
- External Tank Fuel Level

Others:

- Accessory Data Link
- Battery Charger System Voltage
- Battery Voltage
- Digital Output #1
- Digital Output #2
- S/D Override Switch
- Emergency Stop Switch
- Engine in Cool Down
- Engine Failure to Start
- Engine Speed
- Control not in Auto
- Gen Current
- Gen Frequency
- Gen Real Power
- Gen Voltage
- Primary Data Link
- SCADA Data Link
- Maintenance Lamp
- Unexpected Engine Shutdown
- Earth Fault
- Earth Leakage
- Engine Control
- Fuel Leak
- Air Damper Closed
- ATS (Automatic Transfer Switch) in Normal Position
- ATS (Automatic Transfer Switch) in Emergency Position

Others (*continued*):

- Battery Charger Failure
- Gen Breaker Open
- Gen Breaker Closed
- Gen Breaker Lock Out
- Gen Breaker Open Failure
- Gen Breaker Close Failure
- Utility Breaker Open
- Utility Breaker Closed
- Utility Breaker Lock Out
- Utility Breaker Open Failure
- Utility Breaker Close Failure
- Utility Gen Transfer Failure
- Gen Utility Transfer Failure
- Loss of Utility
- Custom Event #1
- Custom Event #2
- Custom Event #3
- Custom Event #4

6. Select the required SPN and press the Enter key to save.

7 Retrofitting PowerWizard

When replacing a PowerWizard or when fitting a new one, ensure that the correct controller type is selected i.e. PowerWizard 1.0 or PowerWizard 2.0. The differences between these are listed in the PowerWizard Variations section. PowerWizard modules should be replaced like for like.

The Electrical Connection drawing for the PowerWizard 1.0 and 2.0 are shown in figure 8 and 9 respectively.

7.1 EST Availability and EST Licensing

The Electronic Service Tool (EST) used with a laptop provides the mechanism for servicing the PowerWizard controller. It allows the user to display, view or change the current Configuration files (Field Replacement Files) or the base level flash files. Therefore EST is an essential service tool for carrying out service operations on the PowerWizard.

Instructions on purchasing and licensing EST can be found on the Dealer website or by contacting the EST Software License and Support Coordinator. Refer to the contact list at the back of this manual.

7.2 Flash Files and Field Replacement Files

PowerWizard have two main types of files associated with them, the flash file (.fls file) that contains the base code and the Field Replacement File (.xml file) that contains the configuration information.

Flash Files:

To obtain a replacement flash file or the most recent version of a flash file, please refer to the contact list at the back of this manual. When enquiring, you will need the Generating Set Serial Number. You will receive the most recent version of the file associated with that Generating Set Serial Number. These files are backward compatible.

A choice of language is available.

If the file is for a new job not associated with a Generating set Serial Number, please contact the After Sales Helpdesk. Refer to the contact list at the back of this manual. The following information is required in order to select the correct flash file for the controller:

1. Controller type (PowerWizard 1.0 or 2.0).
2. After Market ID (AMID).
3. Language.

Field Replacement Files:

The Field Replacement Files only exist for controllers that have previously been programmed at the factory. Field Replacement Files for controllers that are in the field and that have not previously been programmed at the factory, cannot be created by the help desk.

However Field Replacement Files can be used on more than one controller. So if the configuration you want is the same as a previous set, the same Field Replacement File can be used.

Contact the After Sales Helpdesk supplying the Generating Set Serial Number and they will send you the required Field Replacement File. Refer to the contact list at the back of this manual.

Note – If a Field Replacement File that was created for a PowerWizard 1.0 is loaded onto a PowerWizard 2.0, some of the items on the PowerWizard 2.0 will not be configured. To configure these items open EST and select Configuration (SERVICE → CONFIGURATION).

7.2.1 Using EST to load Flash Files

Since new features are continually added, it is strongly recommended to use the latest version of the service tool.

1. Open EST Winflash (This should be a separate option to "Electronic Service Tool" from within the EST sub menu of Windows or it can be accessed from the Service Tool menu by selecting "Utilities" -> "Winflash".
2. Ensure that the service tool is using RP1210 (under "Utilities" -> "preferences" -> "communications") when using a serial connection.
3. Ensure that the communications adapter is connected to the correct port of the PC.
4. You should have a screen similar to:

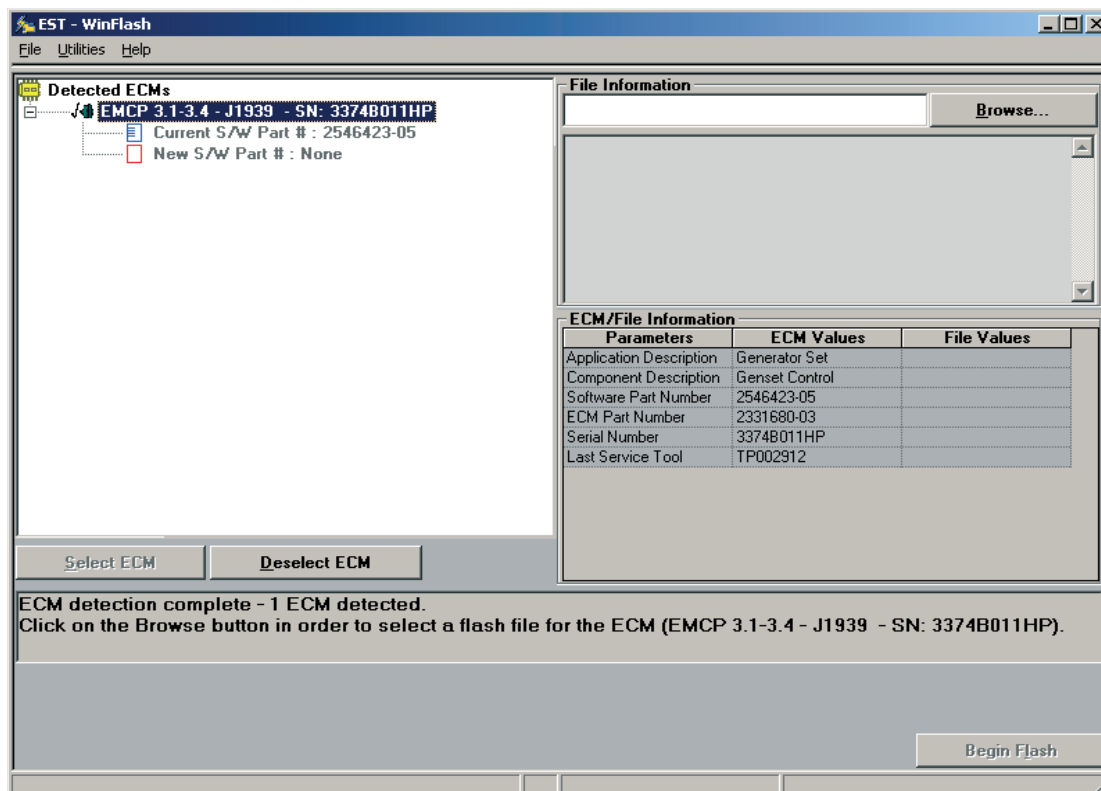


Figure 23: EST Winflash

Press, "Browse" and select the desired flash file

5. Press "Begin Flash"
6. When you see screen below you are finished:



Figure 24: Screen shot after PowerWizard has successfully installed new software

All the (old) setpoints are preserved when the PowerWizard control is re-flashed.

7.2.2 Using EST to load Field Replacement Files.

1. Open EST ECM Replacement by selecting the Service → Copy Configuration → ECM Replacement menu item, as shown in the screen below:

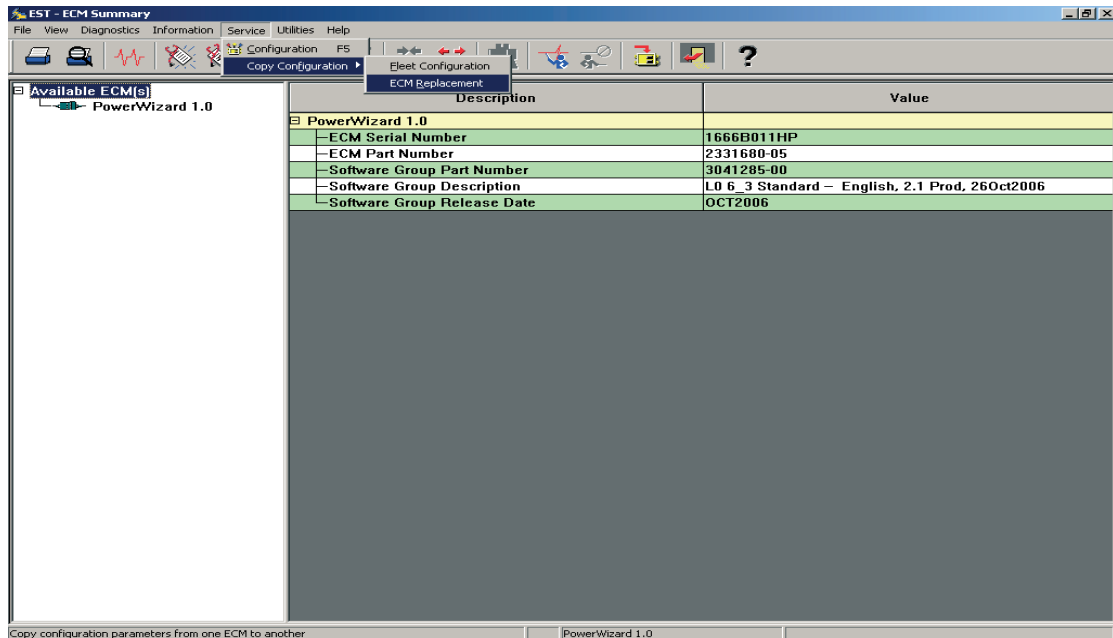


Figure 25: ECM Summary Screen

2. Select the Field replacement File (.xml file) that you want to load as shown in the screen below:

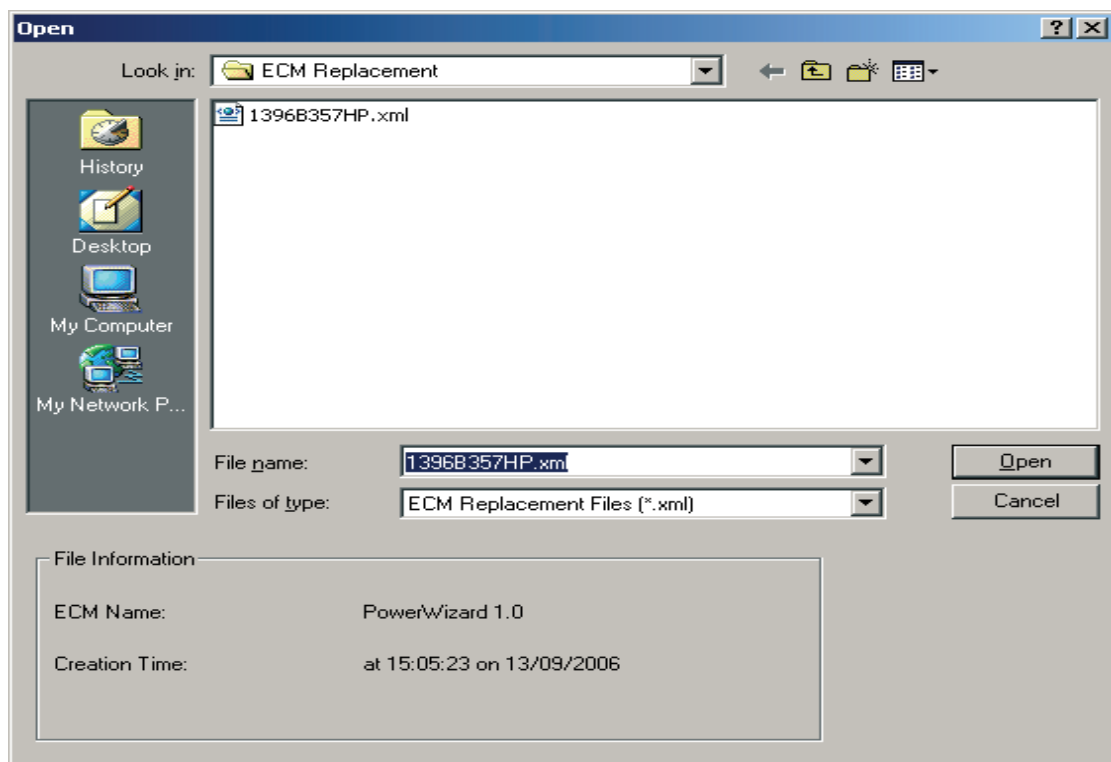


Figure 26: Browse for Field Replacement File

The following screen will then be displayed showing the configuration values:

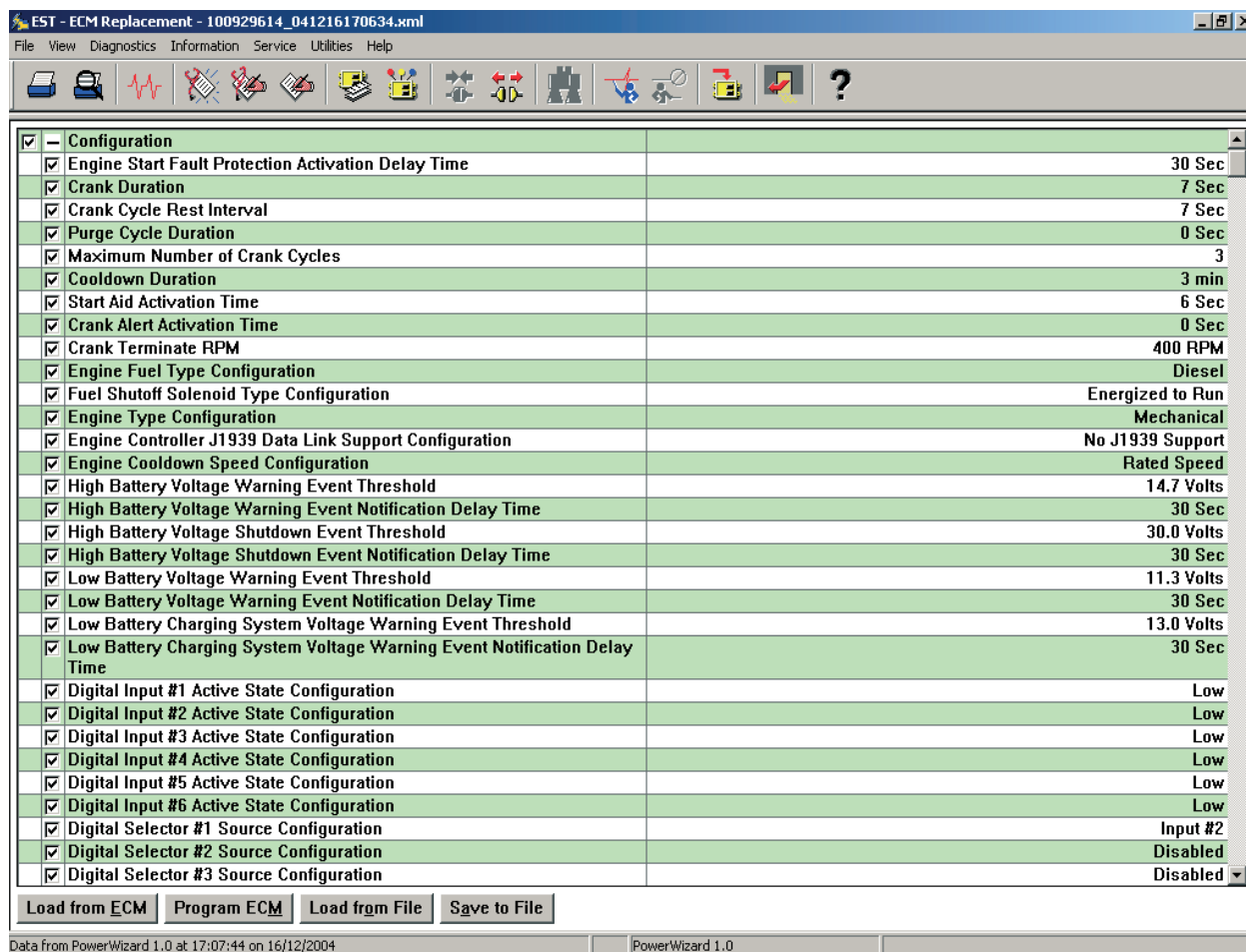


Figure 27: Field Replacement File Configuration Values

3. Select Program ECM to load the configuration values onto the controller. Once the configuration values have been successfully loaded the message "Programming Complete" will be displayed.

7.3 Possible EST Error Messages, their Cause and suggested Action

"The communication adapter was unable to connect to the J1939 data link"

Cause: EST was unable to see the communication adaptor on the port it is expecting it to be connected to:

- Ensure that the communication adapter is connected to the PowerWizard panel and has power
- Ensure that the communication adapter is connected to the PC
- Ensure that the settings on the service tool are set for the correct port
- Ensure the software is configured for the appropriate communication adapter (under utilities -> preferences -> communications).

"No ECMs detected"

Cause: EST was unable to detect any PowerWizard modules on the J1939 network:

- Ensure the PowerWizard module is powered up and is not in Reduced Power Mode
- Check the J1939 wiring between the PowerWizard and the communications adapter, particularly the termination resistor(s).

"The ECM software file and the ECM are not compatible. Process aborted. - Error Code: 163840"

Cause: The flash file is not compatible with the PowerWizard. This is usually seen when trying to flash a PowerWizard 1.0 with a PowerWizard 2.0 flash file (or vice versa):

- Use a flash file that is compatible with the level of PowerWizard control you are trying to flash.

"No flash file selected for the ECM"

Cause: This rather misleading error message is also seen when trying to use a flash file that is not suitable for any level of PowerWizard control:

- Use a flash file that is compatible with the level of PowerWizard control you are trying to flash.

8 Step Through Guides

8.1 Reduced Power Mode (RPM)

Under steady state operation, the PowerWizard controllers on 12V sets have approximately an 850 m Amp current draw (not including any relay loads).

This can be reduced by approximately a factor of 7 using the Reduced Power Mode (RPM). However it is recommended that a generating set at rest or in storage for prolonged periods should either have the battery charger or isolator switch option fitted.

When in reduced power mode, the generating set control will appear powered down. It will also systematically pulse the control panel lamps. In reduced power mode, the control disables all communications.

If Reduced Power Mode is enabled and the generating set is stopped, the control will enter reduced power mode for a preset time (normally 30 minutes) after the last key press. The control will awaken from reduced Power Mode upon a key press, an Emergency Stop or a Remote Start signal.

8.1.1 Enabling Reduced Power Mode Using EST

1. Open EST.
2. Ensure that the service tool is using RP1210 (under "Utilities" -> "preferences" -> "communications").
3. Ensure that the communications adapter is connected to the correct port of the PC.
4. Connect to the PowerWizard control (Press F8).
5. Select Configuration (Service -> Configuration) or (press F5).
6. It will take some time for the service tool to check which blocks are installed. After about 30 seconds the screen will look like this:

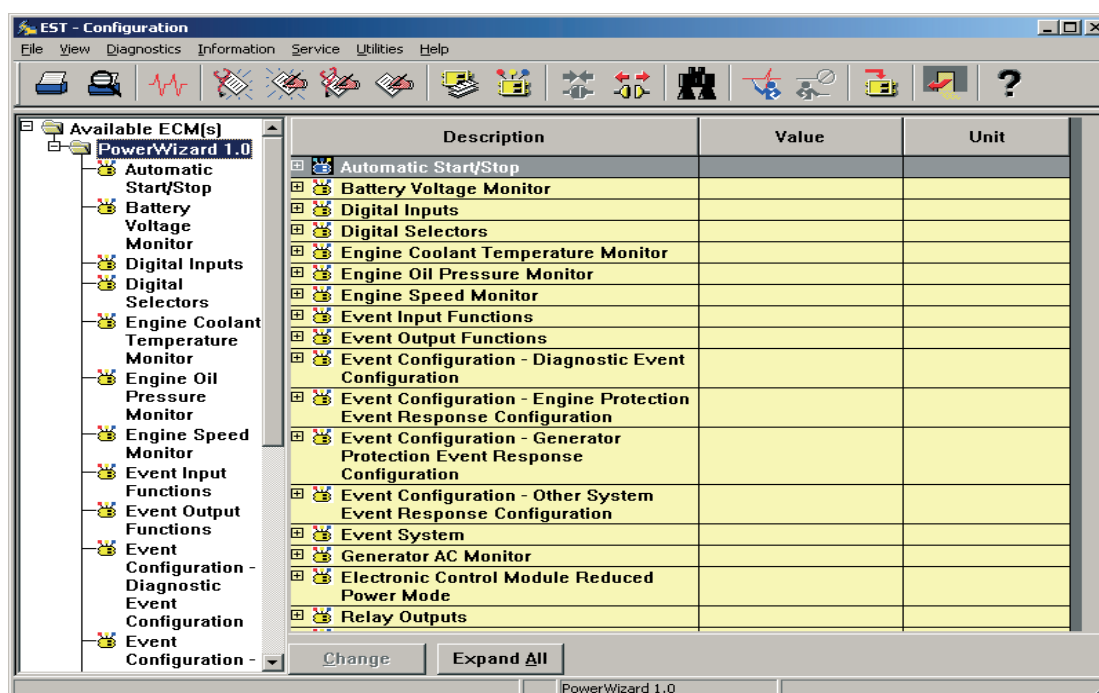


Figure 28: EST Configuration Screen

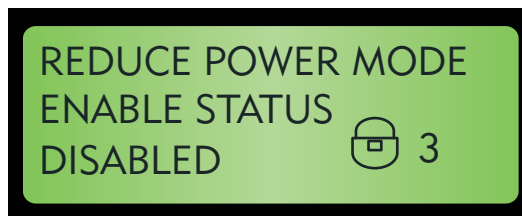
7. Select "Electronic Control Module Reduced Power Mode".
8. At the setpoint "Electronic Control Module Reduced Power Mode Enable Status" click on "Disabled" and change it to "Enabled".
9. Ensure the "Electronic Control Module Reduced Power Mode delay time" is set to an appropriate time (such as 30 minutes).
10. You are finished. Disconnect EST from the PowerWizard.

8.1.2 Enabling Reduced Power Mode Using keypad

To enable the Reduced Power Mode on the PowerWizard by using the keypad go to the Reduce PWR Mode screen as shown below:

CONFIGURATION → SETPOINTS → OTHERS → REDUCE PWR MODE

Enter the following screen:



A Level 3 password is required to allow the user to enter this screen and to enable the Reduce Power Mode. The Level 3 password is obtained by contacting the After Sales Helpdesk. Refer to the contact list at the back of this manual. See also section 4.3-Security.

The user may also change the time delay before Reduced Power Mode is activated, by entering the screen shown below:



Note - The remote communications (RS485) and the J1939 communications will not operate when the controller is in Reduced Power Mode. Therefore if using remote communications ensure Reduced Power Mode is disabled.

Since the remote communications options (MCM7/MCM8) uses a RS485/RS232 converter that is powered via a mains powered charger, we assume that if using remote communications, mains power is available, the controller has a charger and Reduced Power Mode is disabled.

8.2 Service Maintenance Interval

The service maintenance interval is available on PowerWizard 2.0 and has a flash file default of 500 engine hours or 180 days.

8.2.1 Reset Service Interval Alarm

To reset the Service Interval go to the Engine Overview screen and scroll to the Service Interval screen:

SERV OR	-1 HRS 5 DAYS
RESET SERV INTVL?	

Press Enter  to reset the Service Interval.

This resets the hours/days back to their starting position, for example 500 HRS or 180 DAYS.

You may require a level 3 password to do this.

You may use the service tool to change this password level from 3 to 2.

8.2.2 Change Duration of Service Interval Alarm

The Service Interval duration can be set for both days and hours. If the interval in days causes the alarm to be raised before the interval hours run is reached, we suggest that you set the interval in days high (example; set to the maximum of 365 DAYS). The DAYS cannot be disabled.

You must use the service tool (EST) to change the duration of the Service Interval. However the Interval can be viewed by going to the SERV MAINT INTERVAL screen as follows.

CONFIGURATION → SETPOINTS → ENG MON/PROTECT → SERV MAINT INTERVAL

8.2.3 Disable Service Interval Alarm

If the customer does not want this event he can disable it (setting the event response configuration to disabled will stop the event from occurring).

CONFIGURATION → SETPOINTS → EVENTS → EVENT RESPONSE CONFIG → ENGINE PROTECTION EVENTS → OTHER SYSTEM CONFIG → Service Maintenance Interval Warning Event Response Configuration

8.3 Setting Up Language

The Power Wizard module gives the operator the choice of using one of two available languages.

1. Technical English
2. Customer Language

To select your preferred language scroll to the LANGUAGE menu as shown below.

MAIN MENU → PREFERENCES → LANGUAGE

Use the cursor to highlight the preferred language and press the Enter key to select it.

8.4 Disabling NOT IN AUTO

By default PowerWizard control panels have a generating set “not in auto” warning. This warning will be active when the control is in “STOP” or “RUN” mode. For certain applications it may be suitable to disable this warning. To disable “Not in Auto” perform the following.

Go to MAIN MENU → CONFIGURATION → SETPOINTS → EVENTS → EVENT RESPONSE CONFIG → OTHER SYSTEM CONFIG EVENT → OTHERS

Scroll to “Not in Auto”

Select edit by using left cursor key and then press Enter:



This takes you to the screen that is used to configure the desired response for the Not In Auto alarm. The options available can be viewed or edited:

☒	WARNING
☒	AUDIABLE ALERT
☐	SOFT SHUTDOWN
☐	HARD SHUTDOWN
☐	BRK 1 TRIP
☐	BRK 2 TRIP

A ☒ indicates which options are selected. The user may select or remove ☒ by using the left cursor key. If unable to EDIT check that the set is in STOP position and that the required password level is used.

8.5 Disabling Thermo Start (Start Aid Activation)

To disable the Thermo Start go to the “START AID ACTIVATION TIME” screen as shown below:

- CONFIGURATION → SETPOINTS → CONTROL → AUTO START/STOP
- Press enter and then scroll down till you come to the “START AID ACTIVATION TIME” screen.

Once at this screen, set the time to 0 seconds.

9 Contact List

EST Software License and Support Coordinator.

Tel: +44 (0) 28 2826 5228

email: support_tools@fgwilson.com

After Sales Helpdesk

Tel: +44 (0) 28 2826 5001

email: afterSales@fgwilson.com.

(To obtain Flash files/field replacement files, please contact the After Sales Helpdesk)

Please refer to your Secure Dealer Site.

Appendix A – SPN / FMI List

Event Name	SPN	FMI
External Tank High Fuel Level Shutdown	38	0
External Tank Low Fuel Level Shutdown	38	1
External Tank Fuel Level Sensor Short High	38	3
External Tank Fuel Level Sensor Short Low	38	4
External Tank High Fuel Level Warning	38	15
External Tank Low Fuel Level Warning	38	17
Low Starting Air Pressure Shutdown	82	1
Starting Air Pressure Sensor Short High	82	3
Starting Air Pressure Sensor Short Low	82	4
High Starting Air Pressure Warning	82	15
Low Starting Air Pressure Warning	82	17
High Fuel Filter Differential Pressure Shutdown	95	0
Low Fuel Filter Differential Pressure Shutdown	95	1
Fuel Filter Differential Pressure Sensor Short High	95	3
Fuel Filter Differential Pressure Sensor Short Low	95	4
High Fuel Filter Differential Pressure Warning	95	15
Low Fuel Filter Differential Pressure Warning	95	17
High Fuel Level Shutdown	96	0
Low Fuel Level Shutdown	96	1
Fuel Level Sensor Short High	96	3
Fuel Level Sensor Short Low	96	4
High Fuel Level Warning	96	15
Low Fuel Level Warning	96	17
High Engine Oil Level Shutdown	98	0
Low Engine Oil Level Shutdown	98	1
Engine Oil Level Sensor Short High	98	3
Engine Oil Level Sensor Short Low	98	4
High Engine Oil Level Warning	98	15
Low Engine Oil Level Warning	98	17
High Engine Oil Filter Differential Pressure Shutdown	99	0
Low Engine Oil Filter Differential Pressure Shutdown	99	1
Engine Oil Filter Differential Pressure Sensor Short High	99	3
Engine Oil Filter Differential Pressure Sensor Short Low	99	4
High Engine Oil Filter Differential Pressure Warning	99	15
Low Engine Oil Filter Differential Pressure Warning	99	17
Low Engine Oil Pressure Shutdown	100	1
Engine Oil Pressure Sensor Short High	100	3
Engine Oil Pressure Sensor Short Low	100	4
Low Engine Oil Pressure Warning	100	17
High Air Filter Differential Pressure Shutdown	107	0
Low Air Filter Differential Pressure Shutdown	107	1

Event Name	SPN	FMI
Air Filter Differential Pressure Sensor Short High	107	3
Air Filter Differential Pressure Sensor Short Low	107	4
High Air Filter Differential Pressure Warning	107	15
Low Air Filter Differential Pressure Warning	107	17
High Engine Coolant Temperature Shutdown	110	0
Engine Coolant Temperature Sensor Short High	110	3
Engine Coolant Temperature Sensor Short Low	110	4
High Engine Coolant Temperature Warning	110	15
Low Engine Coolant Temperature Warning	110	17
High Engine Coolant Level Shutdown	111	0
Low Engine Coolant Level Shutdown	111	1
Engine Coolant Level Sensor Short High	111	3
Engine Coolant Level Sensor Short Low	111	4
High Engine Coolant Level Warning	111	15
Low Engine Coolant Level Warning	111	17
High Fire Extinguisher Pressure Shutdown	137	0
Low Fire Extinguisher Pressure Shutdown	137	1
Fire Extinguisher Pressure Sensor Short High	137	3
Fire Extinguisher Pressure Sensor Short Low	137	4
High Fire Extinguisher Pressure Warning	137	15
Low Fire Extinguisher Pressure Warning	137	17
Battery Charger Failure	167	11
Battery Charger Failure	167	31
Low Battery Charge System Voltage Warning	167	17
High Battery Voltage Shutdown	168	0
High Battery Voltage Warning	168	15
Low Battery Voltage Warning	168	17
High Ambient Air Temperature Shutdown	171	0
Low Ambient Air Temperature Shutdown	171	1
Ambient Air Temperature Sensor Short High	171	3
Ambient Air Temperature Sensor Short Low	171	4
High Ambient Air Temperature Warning	171	15
Low Ambient Air Temperature Warning	171	17
High Exhaust Temperature Shutdown	173	0
Low Exhaust Temperature Shutdown	173	1
Exhaust Temperature Sensor Short High	173	3
Exhaust Temperature Sensor Short Low	173	4
High Exhaust Temperature Warning	173	15
Low Exhaust Temperature Warning	173	17
High Engine Oil Temperature Shutdown	175	0
Low Engine Oil Temperature Shutdown	175	1
Engine Oil Temperature Sensor Short High	175	3
Engine Oil Temperature Sensor Short Low	175	4
High Engine Oil Temperature Warning	175	15
Low Engine Oil Temperature Warning	175	17

Event Name	SPN	FMI
Engine Over Speed Shutdown	190	0
Engine Under Speed Shutdown	190	1
Engine Speed Sensor Erratic or Not Present	190	2
Engine Speed Sensor Open	190	5
Engine Under Speed Warning	190	17
Modbus Data Link Configuration Error	625	2
Primary Data Link Fault	639	11
Custom Event #1 High Shutdown	701	0
Custom Event #1 Low Shutdown	701	1
Custom Event #1 High Warning	701	15
Custom Event #1 Low Warning	701	17
Custom Event #1 Status	701	31
Custom Event #2 High Shutdown	702	0
Custom Event #2 Low Shutdown	702	1
Custom Event #2 High Warning	702	15
Custom Event #2 Low Warning	702	17
Custom Event #2 Status	702	31
Custom Event #3 High Shutdown	703	0
Custom Event #3 Low Shutdown	703	1
Custom Event #3 High Warning	703	15
Custom Event #3 Low Warning	703	17
Custom Event #3 Status	703	31
Custom Event #4 High Shutdown	704	0
Custom Event #4 Low Shutdown	704	1
Custom Event #4 High Warning	704	15
Custom Event #4 Low Warning	704	17
Custom Event #4 Status	704	31
Custom Event #5 High Shutdown	705	0
Custom Event #5 Low Shutdown	705	1
Custom Event #5 High Warning	705	15
Custom Event #5 Low Warning	705	17
Custom Event #5 Status	705	31
Custom Event #6 High Shutdown	706	0
Custom Event #6 Low Shutdown	706	1
Custom Event #6 High Warning	706	15
Custom Event #6 Low Warning	706	17
Custom Event #6 Status	706	31
Custom Event #7 High Shutdown	707	0
Custom Event #7 Low Shutdown	707	1
Custom Event #7 High Warning	707	15
Custom Event #7 Low Warning	707	17
Custom Event #7 Status	707	31
Custom Event #8 High Shutdown	708	0
Custom Event #8 Low Warning	708	17

Event Name	SPN	FMI
Custom Event #8 Low Shutdown	708	1
Custom Event #8 High Warning	708	15
Custom Event #8 Status	708	31
Custom Event #9 High Shutdown	709	0
Custom Event #9 Low Shutdown	709	1
Custom Event #9 High Warning	709	15
Custom Event #9 Low Warning	709	17
Custom Event #9 Status	709	31
Custom Event #10 High Shutdown	710	0
Custom Event #10 Low Shutdown	710	1
Custom Event #10 High Warning	710	15
Custom Event #10 Low Warning	710	17
Custom Event #10 Status	710	31
Custom Event #11 High Shutdown	711	0
Custom Event #11 Low Shutdown	711	1
Custom Event #11 High Warning	711	15
Custom Event #11 Low Warning	711	17
Custom Event #11 Status	711	31
Custom Event #12 High Shutdown	712	0
Custom Event #12 Low Shutdown	712	1
Custom Event #12 High Warning	712	15
Custom Event #12 Low Warning	712	17
Custom Event #12 Status	712	31
Custom Event #13 High Shutdown	713	0
Custom Event #13 Low Shutdown	713	1
Custom Event #13 High Warning	713	15
Custom Event #13 Low Warning	713	17
Custom Event #13 Status	713	31
Custom Event #14 High Shutdown	714	0
Custom Event #14 Low Shutdown	714	1
Custom Event #14 High Warning	714	15
Custom Event #14 Low Warning	714	17
Custom Event #14 Status	714	31
Custom Event #15 High Shutdown	715	0
Custom Event #15 Low Shutdown	715	1
Custom Event #15 High Warning	715	15
Custom Event #15 Low Warning	715	17
Custom Event #15 Status	715	31
Custom Event #16 High Shutdown	716	0
Custom Event #16 Low Shutdown	716	1
Custom Event #16 High Warning	716	15
Custom Event #16 Low Warning	716	17
Custom Event #16 Status	716	31
Digital Output #1 Short High	924	3
Emergency Stop Switch Activated	970	31

Event Name	SPN	FMI
Digital Output #2 Short High	925	3
Digital Output #3 Short High	926	3
High Generating Set Rear Bearing Temperature Shutdown	1122	0
Low Generating Set Rear Bearing Temperature Shutdown	1122	1
Generating Set Rear Bearing Temperature Sensor Short High	1122	3
Generating Set Rear Bearing Temperature Sensor Short Low	1122	4
High Generating Set Rear Bearing Temperature Warning	1122	15
Low Generating Set Rear Bearing Temperature Warning	1122	17
Accessory Data Link Fault	1231	11
Emergency Shutdown Override Mode Active	1237	31
Fuel Tank Leak	1239	31
Unexpected Engine Shutdown	1383	11
Engine Failure to Start	1664	31
High Right Exhaust Temperature Shutdown	2433	0
Low Right Exhaust Temperature Shutdown	2433	1
Right Exhaust Temperature Sensor Short High	2433	3
Right Exhaust Temperature Sensor Short Low	2433	4
High Right Exhaust Temperature Warning	2433	15
Low Right Exhaust Temperature Warning	2433	17
High Left Exhaust Temperature Warning	2434	0
Low Left Exhaust Temperature Warning	2434	1
Left Exhaust Temperature Sensor Short High	2434	3
Left Exhaust Temperature Sensor Short Low	2434	4
High Left Exhaust Temperature Warning	2434	15
Low Left Exhaust Temperature Warning	2434	17
Generating Set Over Frequency Shutdown	2436	0
Generating Set Under Frequency Shutdown	2436	1
Engine Speed-Generating Set Output Frequency Mismatch Warning	2436	2
Generating Set Output Sensing System Failure	2436	12
Generating Set Over Frequency Warning	2436	15
Generating Set Under Frequency Warning	2436	17
Generating Set Over Voltage Shutdown	2440	0
Generating Set Under Voltage Shutdown	2440	1
Generating Set Over Voltage Warning	2440	15
Generating Set Under Voltage Warning	2440	17
Generating Set Over Current Shutdown	2448	0
Generating Set Over Current Warning	2448	15
Generating Set Reverse Power Shutdown	2452	1
Generating Set Reverse Power Warning	2452	17
Dead Bus Inconsistent Sensing Warning	2530	2
Digital Output #4 Short High	2646	3
Service Maintenance Interval Warning	2648	31
Generating Set Control Not in Automatic Warning	3567	31
Generating Set Breaker Failure to Open	3571	0
Utility Breaker Failure to Close	3574	0

Event Name	SPN	FMI
Utility Breaker Failure to Open	3573	0
Generating Set Breaker Failure to Close	3572	0
Utility to Generating Set Transfer Failure Shutdown	3575	0
Utility to Generating Set Transfer Failure Warning	3575	15
Generating Set to Utility Transfer Failure Warning	3576	15
Loss of Utility	3577	31
Earth Fault	3569	31
Earth Leakage	3570	31
Generating Set to Bus Synchronization Failure Shutdown	3578	0
Generating Set Soft Unload Failure Shutdown	3580	0
Generating Set to Bus Phase Sequence Mismatch Warning	3579	31
Ether Start Relay Open Circuit	3587	5
Ether Start Relay Open Circuit	3588	5
Pre-Lube Relay Open Circuit	3589	5
Ether Hold Relay Short Circuit	3587	6
Ether Start Relay Short Circuit	3588	6
Pre-Lube Relay Short Circuit	3589	6
Generating Set to Bus Synchronization Failure Warning	3578	15
Generating Set Soft Unload Failure Warning	3580	15
SCADA Data Link Fault	3581	11
Air Damper Closed	520198	31
ATS in Normal Position	520232	31
ATS in Emergency Position	520233	31
Generating Set Breaker Closed	520234	31
Utility Breaker Closed	520237	31
Engine in Cooldown	520199	31
Generating Set Circuit Breaker Open	520235	31
Utility Breaker Open	520238	31
Generating Set Breaker Locked Out	520236	31
Utility Breaker Locked Out	520239	31

Appendix B – Setpoints

Setpoints are used in the PowerWizard control panel to tailor the control for each installation. To accommodate the large number of setpoints within PowerWizard the setpoints have been grouped into blocks.

Blocks for Setpoint Configuration

This appendix contains information on the various software blocks in the PowerWizard control. Blocks are functional groups of setpoints. Blocks can occur in multiple instances and in order to program a setpoint, the block number as well as the instance number, must be referenced.

BLOCKS FOR SETPOINT CONFIGURATION			
Block Number	Block Name	PowerWizard 1.0 Number of Instances	PowerWizard 2.0 Number of Instances
1	Engine Oil Pressure Monitor	1	1
4	Engine Coolant Temperature Monitor	1	1
6	Engine Speed Monitor	1	1
7	Battery Voltage Monitor	1	1
8	Engine Operating Hours Monitor	1	1
9	Crank/Attempt Successful Start Counter	1	1
10	Engine Protection Events Configuration	1	1
11	Generating Set Protection Events Configuration	1	1
12	Other System Events Configuration	1	1
17	Diagnostics Configuration	1	1
18	Automatic Start/Stop	1	1
19	Generating Set AC Monitor	1	1
21	Generating Set Over/Under Voltage	1	1
22	Generating Set Over/Under Frequency	1	1
23	Generating Set Over Current	1	1
34	Utility Breaker Control	1	1
35	Generating Set Breaker Control	1	1
53	Digital Inputs	8	8
56	Relay Outputs	8	8
72	Event System	1	1
75	Electronic Control Module Reduced Power Mode	1	1
88	Digital Outputs	1	1
94	Data Link - SCADA	1	1
97	Event Output Functions	30	30
98	Generating Set AC Power Monitor	1	1
100	Service Maintenance	1	1
102	Governor Desired Engine Speed Request	1	1
104	Digital Selectors	10	10
105	Event Input Functions	8	8
106	Spare Analog Input	1	1

SETPOINT INFORMATION FOR SETPOINT CONFIGURATION

Block Number	Setpoint Number	Setpoint Name	Security Level	Min	Max	Units
1	1	Engine Oil Pressure Sensor Configuration	Lvl 3	See Setpoint Value List		
	2	Low Engine Oil Pressure Warning Event Threshold	Lvl 2	34	690	kPa
	3	Low Idle Low Engine Oil Pressure Warning Event Threshold	Lvl 2	34	690	kPa
	4	Low Engine Oil Pressure Warning Event Notification Delay Time	Lvl 2	0	30	seconds
	5	Low Engine Oil Pressure Shutdown Event Threshold	Lvl 2	34	690	kPa
	6	Low Idle Low Engine Oil Pressure Shutdown Event Threshold	Lvl 2	34	690	kPa
	7	Low Engine Oil Pressure Shutdown Event Notification Delay Time	Lvl 2	0	30	seconds
	8	Low Engine Oil Pressure Step Speed	Lvl 2	400	1800	rpm
4	1	Engine Coolant Temperature Sensor Configuration	Lvl 3	See Setpoint Value List		
	2	High Engine Coolant Temperature Warning Event Threshold	Lvl 2	49	120	deg. C
	3	High Engine Coolant Temperature Warning Event Notification Delay Time	Lvl 2	0	30	seconds
	4	High Engine Coolant Temperature Shutdown Event Threshold	Lvl 2	49	120	deg. C
	5	High Engine Coolant Temperature Shutdown Event Notification Delay Time	Lvl 2	0	30	seconds
	6	Low Engine Coolant Temperature Warning Event Threshold	Lvl 2	0	36	deg. C
	7	Low Engine Coolant Temperature Warning Event Notification Delay Time	Lvl 2	0	30	seconds
6	1	Flywheel Teeth	Lvl 3	95	350	N/A
	2	Engine Over Speed Shutdown Event Threshold	Lvl 3	400	4330	rpm
	3	Engine Under Speed Warning Event Threshold	Lvl 3	400	4330	rpm
	4	Engine Under Speed Warning Event Notification Delay Time	Lvl 2	0	20	seconds
	5	Engine Under Speed Shutdown Event Threshold	Lvl 3	400	4330	rpm
	6	Engine Under Speed Shutdown Event Notification Delay Time	Lvl 2	0	20	seconds
	7	Engine Speed Sensor Configuration	Svc Tool	See Setpoint Value List		
7	1	High Battery Voltage Warning Event Threshold	Lvl 1	12	50	VDC
	2	High Battery Voltage Warning Event Notification Delay Time	Lvl 1	0	240	seconds
	3	High Battery Voltage Shutdown Event Threshold	Lvl 1	12	50	VDC
	4	High Battery Voltage Shutdown Event Notification Delay Time	Lvl 1	0	240	seconds
	5	Low Battery Voltage Warning Event Threshold	Lvl 1	0	25	VDC
	6	Low Battery Voltage Warning Event Notification Delay Time	Lvl 1	0	240	seconds
	7	Low Battery Charging System Voltage Warning Event Threshold	Lvl 1	0	30	VDC
	8	Low Battery Charging System Voltage Warning Event Notification Delay Time	Lvl 1	0	240	seconds
9	1	Customer Password Security Level to Reset Crank/Start Counters	Lvl 3	0	4	N/A
10	1	Air Damper Closed Event Response Configuration	Lvl 2	See Response Configuration Options		
	2	High Air Filter Differential Pressure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	3	High Air Filter Differential Pressure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	4	Low Air Filter Differential Pressure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	5	Low Air Filter Differential Pressure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	6	Emergency Stop Activated Event Response Configuration	Lvl 2	See Response Configuration Options		

SETPOINT INFORMATION FOR SETPOINT CONFIGURATION

Block Number	Setpoint Number	Setpoint Name	Security Level	Min	Max	Units
	7	High Engine Coolant Level Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	8	High Engine Coolant Level Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	9	Low Engine Coolant Level Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	10	Low Engine Coolant Level Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	11	High Engine Coolant Temperature Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	12	High Engine Coolant Temperature Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	13	Low Engine Coolant Temperature Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	14	Engine Failure to Start Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	15	High Engine Oil Level Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	16	High Engine Oil Level Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	17	Low Engine Oil Level Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	18	Low Engine Oil Level Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	19	Low Engine Oil Pressure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	20	Low Engine Oil Pressure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	21	High Engine Oil Temperature Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	22	High Engine Oil Temperature Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	23	Low Engine Oil Temperature Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	24	Low Engine Oil Temperature Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	25	Unexpected Engine Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	26	Engine Over Speed Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	27	Engine Under Speed Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	28	Engine Under Speed Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	29	High Exhaust Temperature Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	30	High Exhaust Temperature Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	31	Low Exhaust Temperature Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	32	Low Exhaust Temperature Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	33	High Right Exhaust Temperature Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	34	High Right Exhaust Temperature Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		

SETPOINT INFORMATION FOR SETPOINT CONFIGURATION

Block Number	Setpoint Number	Setpoint Name	Security Level	Min	Max	Units
	35	Low Right Exhaust Temperature Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	36	Low Right Exhaust Temperature Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	37	High Left Exhaust Temperature Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	38	High Left Exhaust Temperature Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	39	Low Left Exhaust Temperature Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	40	Low Left Exhaust Temperature Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	41	High Fuel Filter Differential Pressure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	42	High Fuel Filter Differential Pressure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	43	Low Fuel Filter Differential Pressure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	44	Low Fuel Filter Differential Pressure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	45	High Fuel Level Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	46	High Fuel Level Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	47	Low Fuel Level Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	48	Low Fuel Level Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	49	External Tank High Fuel Level Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	50	External Tank High Fuel Level Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	51	External Tank Low Fuel Level Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	52	External Tank Low Fuel Level Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	53	Fuel Tank Leak Event Response Configuration	Lvl 2	See Response Configuration Options		
	54	High Engine Oil Filter Differential Pressure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	55	High Engine Oil Filter Differential Pressure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	56	Low Engine Oil Filter Differential Pressure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	57	Low Engine Oil Filter Differential Pressure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	58	Service Maintenance Interval Warning Event Response Configuration	Svc Tool	See Response Configuration Options		
	59	High Starting Air Pressure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	60	High Starting Air Pressure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	61	Low Starting Air Pressure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	62	Low Starting Air Pressure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	63	High Gas Pressure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		

SETPOINT INFORMATION FOR SETPOINT CONFIGURATION

Block Number	Setpoint Number	Setpoint Name	Security Level	Min	Max	Units
11	64	High Gas Pressure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	65	Low Gas Pressure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	66	Low Gas Pressure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	1	High Generating Set Bearing #1 Temperature Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	2	High Generating Set Bearing #1 Temperature Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	3	Low Generating Set Bearing #1 Temperature Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	4	Low Generating Set Bearing #1 Temperature Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	5	Generating Set Over Current (Amp) Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	6	Generating Set Over Current (Amp) Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	7	Generating Set Over Frequency Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	8	Generating Set Over Frequency Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	9	Generating Set Under Frequency Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	10	Generating Set Under Frequency Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	11	Generating Set Reverse Power Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	12	Generating Set Reverse Power Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	13	Generating Set Over Voltage Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	14	Generating Set Over Voltage Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	15	Generating Set Under Voltage Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	16	Generating Set Under Voltage Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	17	Earth Fault Event Response Configuration	Lvl 2	See Response Configuration Options		
12	1	High Ambient Air Temperature Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	2	High Ambient Air Temperature Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	3	Low Ambient Air Temperature Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	4	Low Ambient Air Temperature Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	5	Automatic Transfer Switch in Normal Position Event Response Configuration	Lvl 2	See Response Configuration Options		
	6	Automatic Transfer Switch in Emergency Position Event Response Configuration	Lvl 2	See Response Configuration Options		
	7	High Battery Voltage Warning Event Response Configuration	Lvl 1	See Response Configuration Options		
	8	High Battery Voltage Shutdown Event Response Configuration	Lvl 1	See Response Configuration Options		
	9	Low Battery Voltage Warning Event Response Configuration	Lvl 1	See Response Configuration Options		

SETPOINT INFORMATION FOR SETPOINT CONFIGURATION

Block Number	Setpoint Number	Setpoint Name	Security Level	Min	Max	Units
	10	Low Battery Charging System Voltage Warning Event Response Configuration	Lvl 1	See Response Configuration Options		
	11	Generating Set Breaker Failure to Open Event Response Configuration	Lvl 3	See Response Configuration Options		
	12	Generating Set Breaker Failure to Close Event Response Configuration	Lvl 3	See Response Configuration Options		
	13	Generating Set Breaker Open Event Response Configuration	Lvl 1	See Response Configuration Options		
	14	Generating Set Breaker Closed Event Response Configuration	Lvl 1	See Response Configuration Options		
	15	Utility Breaker Failure to Open Event Response Configuration	Lvl 3	See Response Configuration Options		
	16	Utility Breaker Failure to Close Event Response Configuration	Lvl 3	See Response Configuration Options		
	17	Utility Breaker Open Event Response Configuration	Lvl 1	See Response Configuration Options		
	18	Utility Breaker Closed Event Response Configuration	Lvl 1	See Response Configuration Options		
	19	Emergency Shutdown Override Mode Active Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	20	Engine in Cooldown Event Response Configuration	Lvl 1	See Response Configuration Options		
	21	Engine Speed-Generating Set Output Frequency Mismatch Event Response Configuration	Lvl 2	See Response Configuration Options		
	22	Custom Event #1 High Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	23	Custom Event #1 High Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	24	Custom Event #1 Low Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	25	Custom Event #1 Low Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	26	Custom Event #1 Event Response Configuration	Lvl 2	See Response Configuration Options		
	27	Custom Event #2 High Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	28	Custom Event #2 High Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	29	Custom Event #2 Low Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	30	Custom Event #2 Low Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	31	Custom Event #2 Event Response Configuration	Lvl 2	See Response Configuration Options		
	32	Custom Event #3 High Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	33	Custom Event #3 High Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	34	Custom Event #3 Low Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	35	Custom Event #3 Low Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	36	Custom Event #3 Event Response Configuration	Lvl 2	See Response Configuration Options		
	37	Custom Event #4 High Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	38	Custom Event #4 High Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		

SETPOINT INFORMATION FOR SETPOINT CONFIGURATION

Block Number	Setpoint Number	Setpoint Name	Security Level	Min	Max	Units
	39	Custom Event #4 Low Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	40	Custom Event #4 Low Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	41	Custom Event #4 Event Response Configuration	Lvl 1	See Response Configuration Options		
	42	Custom Event #5 High Warning Event Response Configuration	Lvl 1	See Response Configuration Options		
	43	Custom Event #5 High Shutdown Event Response Configuration	Lvl 1	See Response Configuration Options		
	44	Custom Event #5 Low Warning Event Response Configuration	Lvl 1	See Response Configuration Options		
	45	Custom Event #5 Low Shutdown Event Response Configuration	Lvl 3	See Response Configuration Options		
	46	Custom Event #5 Event Response Configuration	Lvl 3	See Response Configuration Options		
	47	Custom Event #6 High Warning Event Response Configuration	Lvl 1	See Response Configuration Options		
	48	Custom Event #6 High Shutdown Event Response Configuration	Lvl 1	See Response Configuration Options		
	49	Custom Event #6 Low Warning Event Response Configuration	Lvl 3	See Response Configuration Options		
	50	Custom Event #6 Low Shutdown Event Response Configuration	Lvl 3	See Response Configuration Options		
	51	Custom Event #6 Event Response Configuration	Lvl 1	See Response Configuration Options		
	52	Custom Event #7 High Warning Event Response Configuration	Lvl 1	See Response Configuration Options		
	53	Custom Event #7 High Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	54	Custom Event #7 Low Warning Event Response Configuration	Lvl 1	See Response Configuration Options		
	55	Custom Event #7 Low Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	56	Custom Event #7 Event Response Configuration	Lvl 2	See Response Configuration Options		
	57	Custom Event #8 High Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	58	Custom Event #8 High Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	59	Custom Event #8 Low Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	60	Custom Event #8 Low Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	61	Custom Event #8 Event Response Configuration	Lvl 2	See Response Configuration Options		
	62	High Fire Extinguisher Pressure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	63	High Fire Extinguisher Pressure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	64	Low Fire Extinguisher Pressure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	65	Low Fire Extinguisher Pressure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	66	Generating Set Control Not in Automatic Warning Event Response Configuration	Lvl 1	See Response Configuration Options		
	67	Loss of Utility Event Response Configuration	Lvl 1	See Response Configuration Options		

SETPOINT INFORMATION FOR SETPOINT CONFIGURATION

Block Number	Setpoint Number	Setpoint Name	Security Level	Min	Max	Units
	68	Utility to Generating Set Transfer Failure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	69	Utility to Generating Set Transfer Failure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	70	Generating Set to Utility Transfer Failure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	71	Dead Bus Inconsistent Sensing Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	72	Generating Set to Bus Synchronization Failure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	73	Generating Set to Bus Synchronization Failure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	74	Generating Set to Bus Phase Sequence Mismatch Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	75	Generating Set Not Ready to Automatically Parallel Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	76	Generating Set Soft Unload Failure Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	77	Generating Set Soft Unload Failure Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	78	Custom Event #9 High Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	79	Custom Event #9 High Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	80	Custom Event #9 Low Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	81	Custom Event #9 Low Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	82	Custom Event #9 Event Response Configuration	Lvl 2	See Response Configuration Options		
	83	Custom Event #10 High Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	84	Custom Event #10 High Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	85	Custom Event #10 Low Warning Response Configuration	Lvl 2	See Response Configuration Options		
	86	Custom Event #10 Low Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	87	Custom Event #10 Event Response Configuration	Lvl 2	See Response Configuration Options		
	88	Custom Event #11 High Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	89	Custom Event #11 High Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	90	Custom Event #11 Low Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	91	Custom Event #11 Low Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	92	Custom Event #11 Event Response Configuration	Lvl 2	See Response Configuration Options		
	93	Custom Event #12 High Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	94	Custom Event #12 High Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	95	Custom Event #12 Low Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	96	Custom Event #12 Low Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		

SETPPOINT INFORMATION FOR SETPOINT CONFIGURATION

Block Number	Setpoint Number	Setpoint Name	Security Level	Min	Max	Units
	97	Custom Event #12 Event Response Configuration	Lvl 2	See Response Configuration Options		
	98	Custom Event #13 High Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	99	Custom Event #13 High Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	100	Custom Event #13 Low Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	101	Custom Event #13 Low Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	102	Custom Event #13 Event Response Configuration	Lvl 2	See Response Configuration Options		
	103	Custom Event #14 High Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	104	Custom Event #14 High Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	105	Custom Event #14 Low Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	106	Custom Event #14 Low Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	107	Custom Event #14 Event Response Configuration	Lvl 2	See Response Configuration Options		
	108	Custom Event #15 High Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	109	Custom Event #15 High Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	110	Custom Event #15 Low Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	111	Custom Event #15 Low Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	112	Custom Event #15 Event Response Configuration	Lvl 2	See Response Configuration Options		
	113	Custom Event #16 High Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	114	Custom Event #16 High Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	115	Custom Event #16 Low Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	116	Custom Event #16 Low Shutdown Event Response Configuration	Lvl 2	See Response Configuration Options		
	117	Custom Event #16 Event Response Configuration	Lvl 2	See Response Configuration Options		
	118	Generating Set Breaker Locked Out Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	119	Utility Breaker Locked Out Warning Event Response Configuration	Lvl 2	See Response Configuration Options		
	120	Earth Leakage Event Response Configuration	Lvl 2	See Response Configuration Options		
17	1	Accessory Data Link Diagnostic Response Configuration	Svc Tool	See Response Configuration Options		
	2	Air Filter Differential Pressure Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	3	Ambient Air Temperature Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	4	Digital Output #1 Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	5	Digital Output #2 Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		

SETPOINT INFORMATION FOR SETPOINT CONFIGURATION						
Block Number	Setpoint Number	Setpoint Name	Security Level	Min	Max	Units
	6	Engine Coolant Level Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	7	Engine Coolant Temperature Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	8	Engine Oil Level Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	9	Engine Oil Pressure Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	10	Engine Oil Temperature Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	11	Engine Speed Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	12	Exhaust Temperature Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	13	Right Exhaust Temperature Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	14	Left Exhaust Temperature Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	15	Fire Extinguisher Pressure Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	16	Fuel Filter Differential Pressure Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	17	Fuel Level Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	18	External Tank Fuel Level Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	19	Generating Set Bearing #1 Temperature Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	20	Generating Set Output Sensing System Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	21	Engine Oil Filter Differential Pressure Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	22	Primary Data Link Diagnostic Response Configuration	Lvl 3	See Response Configuration Options		
	23	SCADA Data Link Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	24	Starting Air Pressure Sensor Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	25	Battery Charger Failure Diagnostic Response Configuration	Lvl 1	See Response Configuration Options		
	26	Multiple Generating Set Control Data Link Communication Failure Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	27	Multiple Generating Set Control Data Link Configuration Error Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	28	Control is Offline Multiple Generating Set Control Data Link Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	29	Digital Output #3 Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	30	Digital Output #4 Diagnostic Response Configuration	Lvl 2	See Response Configuration Options		
	31	Engine Controller Not Responding Diagnostic Response Configuration	Lvl 3	See Response Configuration Options		
18	1	Engine Start Fault Protection Activation Delay Time	Lvl 2	0	300	seconds
	2	Crank Duration	Lvl 2	5	300	seconds
	3	Crank Cycle Rest Interval	Lvl 2	5	300	seconds
	4	Engine Purge Cycle Time	Svc Tool	0	20	seconds
	5	Maximum Number of Crank Cycles	Lvl 2	1	20	N/A

SETPOINT INFORMATION FOR SETPOINT CONFIGURATION

Block Number	Setpoint Number	Setpoint Name	Security Level	Min	Max	Units
	6	Cooldown Duration	Lvl 2	0	30	minutes
	7	Start Aid Activation Time	Lvl 2	0	240	seconds
	8	Crank Alert Activation Time	Lvl 1	0	60	seconds
	9	Crank Terminate RPM	Lvl 3	100	1000	rpm
	10	Engine Fuel Type Configuration	Svc Tool	See Setpoint Value List		
	11	Fuel Shutoff Solenoid Type Configuration	Svc Tool	See Setpoint Value List		
	12	Engine Type Configuration	Svc Tool	See Setpoint Value List		
	13	Engine Controller J1939 Data Link Support Configuration	Svc Tool	See Setpoint Value List		
	14	Engine Cooldown Speed Configuration	Lvl 2	See Setpoint Value List		
	15	Engine Operating State Input Configuration	Svc Tool	See Setpoint Value List		
19	1	Generating Set Connection Configuration	Lvl 3	See Setpoint Value List		
	2	Generating Set Potential Transformer Primary Winding Rating	Lvl 3	1	50000	V
	3	Generating Set Potential Transformer Secondary Winding Rating	Lvl 3	1	240	V
	4	Generating Set Current Transformer Primary Winding Rating	Lvl 3	1	7000	A
	5	Generating Set Current Transformer Secondary Winding Rating	Lvl 3	1	5	A
	6	Number of Generating Set Poles	Svc Tool	0	200	N/A
	7	Generating Set Rated Frequency	Lvl 3	See Setpoint Value List		
	8	Generating Set Rated Voltage	Lvl 2	100	50000	V
	9	Generating Set Rated Power	Lvl 3	1	50000	kW
	10	Generating Set Rated Apparent Power	Lvl 3	1	50000	kVA
21	1	Generating Set Over Voltage Warning Event Percentage Threshold	Lvl 2	100	125	%
	2	Generating Set Over Voltage Warning Event Notification Delay Time	Lvl 2	0	120	seconds
	3	Generating Set Over Voltage Shutdown Event Percentage Threshold	Lvl 2	100	125	%
	4	Generating Set Over Voltage Shutdown Event Notification Delay Time	Lvl 2	0	120	seconds
	5	Generating Set Under Voltage Warning Event Percentage Threshold	Lvl 2	60	100	%
	6	Generating Set Under Voltage Warning Event Notification Delay Time	Lvl 2	0	120	seconds
	7	Generating Set Under Voltage Shutdown Event Threshold	Lvl 2	60	100	%
	8	Generating Set Under Voltage Shutdown Event Notification Delay Time	Lvl 2	0	120	seconds
22	1	Generating Set Over Frequency Warning Event Percentage Threshold	Lvl 2	80	120	%
	2	Generating Set Over Frequency Warning Event Notification Delay Time	Lvl 2	0	120	seconds
	3	Generating Set Over Frequency Shutdown Event Percentage Threshold	Lvl 2	80	120	%
	4	Generating Set Over Frequency Shutdown Event Notification Delay Time	Lvl 2	0	120	seconds
	5	Generating Set Under Frequency Warning Event Percentage Threshold	Lvl 2	80	120	%
	6	Generating Set Under Frequency Warning Event Notification Delay Time	Lvl 2	0	120	seconds
	7	Generating Set Under Frequency Shutdown Event Percentage Threshold	Lvl 2	80	120	%
	8	Generating Set Under Frequency Shutdown Event Notification Delay Time	Lvl 2	0	120	seconds
23	1	Generating Set Definite Time Over Current (Amp) Warning Event Percentage Threshold	Lvl 2	80	130	%
	2	Generating Set Inverse Time Over Current (Amp) Shutdown Event Time Multiplier	Lvl 2	0.05	10	seconds
	3	Generating Set Definite Time Over Current (Amp) Shutdown Event Percentage Threshold	Lvl 2	100	300	%
	4	Generating Set Definite Time Over Current (Amp) Shutdown Event Notification Delay Time	Lvl 2	0.1	20	seconds
24	1	Generating Set Reverse Power Warning Event Percentage Threshold	Lvl 2	1	20	%

SETPOINT INFORMATION FOR SETPOINT CONFIGURATION

Block Number	Setpoint Number	Setpoint Name	Security Level	Min	Max	Units
34	2	Generating Set Reverse Power Warning Event Notification Delay Time	Lvl 2	0	30	seconds
	3	Generating Set Reverse Power Shutdown Event Percentage Threshold	Lvl 2	1	20	%
	4	Generating Set Reverse Power Shutdown Event Notification Delay Time	Lvl 2	0	30	seconds
	1	Utility Breaker Closing Pulse Active Time	Lvl 3	0.1	10	seconds
	2	Utility Breaker Closing Pulse Rest Interval	Lvl 3	0	60	seconds
	3	Utility Breaker Maximum Closing Time	Lvl 3	1	120	seconds
	4	Utility Breaker Maximum Opening Time	Lvl 3	1	20	seconds
35	5	Utility Breaker Lockout Configuration	Lvl 2	See Setpoint Value List		
	6	Utility Breaker Type Configuration	Lvl 3	See Setpoint Value List		
	7	Automatic Utility Breaker Control Source Configuration	Lvl 3	See Setpoint Value List		
	1	Generating Set Breaker Closing Pulse Active Time	Lvl 3	0.1	10	seconds
	2	Generating Set Breaker Closing Pulse Rest Interval	Lvl 3	0	60	seconds
	3	Generating Set Breaker Maximum Closing Time	Lvl 3	1	120	seconds
	4	Generating Set Breaker Maximum Opening Time	Lvl 3	1	20	seconds
53	5	Generating Set Breaker Lockout Configuration	Lvl 3	See Setpoint Value List		
	6	Generating Set Breaker Type Configuration	Lvl 3	See Setpoint Value List		
	7	Automatic Generating Set Breaker Control Source Configuration	Lvl 3	See Setpoint Value List		
56	1	Digital Input #NN Active State Configuration	Lvl 2	See Setpoint Value List		
72	1	Relay Output #N Active State Configuration	read-only	See Setpoint Value List		
72	1	Event Warning Condition Response Auto Reset Enable Status	Svc Tool	See Setpoint Value List		
	2	Event Audible Alert Response Auto Reset Enable Status	Svc Tool	See Setpoint Value List		
	3	Event Loss of Utility Response Auto Reset Enable Status	Svc Tool	See Setpoint Value List		
	4	Event Breaker #1 Trip Response Auto Reset Enable Status	Svc Tool	See Setpoint Value List		
	5	Event Breaker #2 Trip Response Auto Reset Enable Status	Svc Tool	See Setpoint Value List		
75	1	Electronic Control Module Reduced Power Mode Enable Status	Lvl 3	See Setpoint Value List		
	2	Electronic Control Module Reduced Power Mode Delay Time	Lvl 2	1	120	minutes
88	1	Digital Output #N Active State Configuration	Lvl 2	See Setpoint Value List		
94	1	SCADA Data Link Baud Rate	Lvl 2	See Setpoint Value List		
	2	SCADA Data Link Parity	Lvl 2	See Setpoint Value List		
	3	SCADA Data Link Slave Address	Lvl 2	1	247	N/A
	4	SCADA Data Link Access Password	Lvl 2	0	0xffffffff	N/A
	5	SCADA Data Link Connection Timeout Interval	Lvl 2	0.1	3600	seconds
	6	RS-485 Bias Resistor Enable Status	Lvl 2	See Setpoint Value List		
96	1	Engine Cylinder Temperature Sensor Installation Status	Lvl 2	See Setpoint Value List		
	2	Number of Engine Cylinders	Lvl 2	1	20	N/A
97	1	Event Output Function #NN Trigger Condition	Lvl 2	See Setpoint Value List		
	2	Event Output Function #NN Suspect Parameter Number	Lvl 2	See Setpoint Value List		
98	1	Customer Password Security Level to Reset Generating Set Energy Meters	Lvl 3	0	3	N/A
100	1	Service Maintenance Interval Hours	Svc Tool	0	2000	hours
	2	Service Maintenance Interval Days	Svc Tool	0	365	days
	3	Customer Password Security Level to Reset Service Maintenance Interval	Lvl 3	0	4	N/A
102	1	Maximum Engine Speed Bias	Lvl 2	0	400	rpm
	2	Generating Set Nominal Output Frequency	Lvl 2	40	500	Hz

SETPOINT INFORMATION FOR SETPOINT CONFIGURATION

Block Number	Setpoint Number	Setpoint Name	Security Level	Min	Max	Units
104	1	Digital Selector #NN Source Configuration	Lvl 2	See Setpoint Value List		
105	1	Event Input Function #NN Active State Configuration	Lvl 2	See Setpoint Value List		
	2	Event Input Function #NN Event Notification Delay Time	Lvl 2	0	250	seconds
	3	Event Input Function #NN Suspect Parameter Number	Lvl 2	See Setpoint Value List		
	4	Event Input Function #NN Failure Mode Identifier	Lvl 2	See Setpoint Value List		
106	1	Spare Analog Input Enable Status	Lvl 2	See Setpoint Value List		
	2	Spare Analog Input Type Configuration	read-only	See Setpoint Value List		
	3	Spare Analog Input Suspect Parameter Number	Lvl 2	See Setpoint Value List		
	4	Spare Analog Input High Percentage Warning Event Threshold	Lvl 2	0	100	%
	5	Spare Analog Input High Temperature Warning Event	Lvl 2	-273	1735	deg. C
	6	Spare Analog Input High Pressure Warning Event Threshold	Lvl 2	-250	10000	kPa
	7	Spare Analog Input High Warning Event Notification Delay Time	Lvl 2	0	60	seconds
	8	Spare Analog Input High Percentage Shutdown Event Threshold	Lvl 3	0	100	%
	9	Spare Analog Input High Temperature Shutdown Event Threshold	Lvl 2	-273	1735	deg. C
	10	Spare Analog Input High Pressure Shutdown Event Threshold	Lvl 2	-250	10000	kPa
	11	Spare Analog Input High Shutdown Event Notification Delay Time	Lvl 2	0	60	seconds
	12	Spare Analog Input Low Percentage Warning Event Threshold	Lvl 2	0	100	%
	13	Spare Analog Input Low Temperature Warning Event Threshold	Lvl 2	-273	1735	deg. C
	14	Spare Analog Input Low Pressure Warning Event Threshold	Lvl 2	-250	10000	kPa
	15	Spare Analog Input Low Warning Event Notification Delay Time	Lvl 2	0	60	seconds
	16	Spare Analog Input Low Percentage Shutdown Event Threshold	Lvl 2	0	100	%
	17	Spare Analog Input Low Temperature Shutdown Event Threshold	Lvl 2	-273	1735	deg. C
	18	Spare Analog Input Low Pressure Shutdown Event Threshold	Lvl 2	-250	10000	kPa
	19	Spare Analog Input Low Shutdown Event Notification Delay Time	Lvl 2	0	60	seconds
108	1	Maximum Generating Set Voltage Output Bias Percentage	Lvl 2	0	100	%
	2	Generating Set Nominal Output Voltage	Lvl 2	100	50000	V
109	1	Generating Set Winding Temperature Sensor Installation Status	Lvl 2	See Setpoint Value List		
	2	Generating Set Bearing Temperature Sensor Installation Configuration	Lvl 2	See Setpoint Value List		

Setpoint Value Lists

Some setpoints above refer to Setpoint Value Lists. Those setpoints are programmed with integer values with each integer having a specific definition. These integer values and representations are given overleaf.

SETPOINT VALUE LISTS

Block Number	Setpoint Number	Setpoint Name	Value	Value Name
1	1	Engine Oil Pressure Sensor Configuration	0	Sensor
			1	Data Link
4	1	Engine Coolant Temperature Sensor Configuration	0	Sensor
			1	Data Link
6	7	Engine Speed Sensor Configuration	0	Sensor
18	10	Engine Fuel Type Configuration	0	Diesel
			1	Natural Gas
	11	Fuel Shutoff Solenoid Type Configuration	0	Energized to Run
			1	Energized to Stop
	12	Engine Type Configuration	0	Mechanical
			1	Electronic
	13	Engine Controller J1939 Data Link Support Configuration	0	No J1939 Support
			1	Basic J1939 Support
			2	Enhanced J1939 Support
	14	Engine Cooldown Speed Configuration	0	Rated Speed
			1	Low Idle
	15	Engine Operating State Input Configuration	0	Hard Wired Input
			1	CAN Input
19	1	Generating Set Connection Configuration	0	Wye (or Star)
			1	Delta (3-Wire)
			2	Delta (4-Wire)
			3	Single Phase (2-Wire)
			4	Single Phase (3-Wire)
	7	Generating Set Rated Frequency	0	50 Hz
			1	60 Hz
			2	400 Hz
34	5	Utility Breaker Lockout Configuration	0	Not Locked Out
			1	Locked Out
	6	Utility Breaker Type Configuration	0	Not Installed
			1	Manual
			2	Automatic
	7	Automatic Utility Breaker Control Source Configuration	0	Supervisory Controlled
			1	Externally Controlled
35	5	Generating Set Breaker Lockout Configuration	0	Not Locked Out
			1	Locked Out
	6	Generating Set Breaker Type Configuration	0	Manual
			1	Automatic
	7	Automatic Generating Set Breaker Control Source Configuration	0	Generating Set Controlled
			1	Externally Controlled
53	1	Digital Input #NN Active State Configuration	0	Low
			1	High
56	1	Relay Output #N Active State Configuration	0	Low
			1	High

SETPOINT VALUE LISTS

Block Number	Setpoint Number	Setpoint Name	Value	Value Name
72	1	Event Warning Condition Response Auto Reset Enable Status	0	Disabled
			1	Enabled
	2	Event Audible Alert Response Auto Reset Enable Status	0	Disabled
			1	Enabled
	3	Event Loss of Utility Response Auto Reset Enable Status	0	Disabled
			1	Enabled
	4	Event Breaker #1 Trip Response Auto Reset Enable Status	0	Disabled
			1	Enabled
	5	Event Breaker #2 Trip Response Auto Reset Enable Status	0	Disabled
			1	Enabled
75	1	Electronic Control Module Reduced Power Mode Enable Status	0	Disabled
			1	Enabled
88	1	Digital Output #N Active State Configuration	0	Low
			1	High
94	1	SCADA Data Link Baud Rate	0	2400 baud
			1	4800 baud
			2	9600 baud
			3	14400 baud
			4	19200 baud
			5	28800 baud
			6	38400 baud
			7	57600 baud
			8	115200 baud
	2	SCADA Data Link Parity	0	None
			1	Odd
	6	RS-485 Bias Resistor Enable Status	0	Disabled
			1	Enabled
96	1	Engine Cylinder Temperature Sensor Installation Status	0	Not Installed
			1	Installed
97	1	Event Output Function #NN Trigger Condition	0	Disabled
			8320	General Event - High or Low Warning
			8352	General Event - High or Low Shutdown
			8448	General Event - High or Low Warning or High or Low Shutdown
			8480	General Event - High or Low Warning or High or Low Shutdown or Diagnostic
			8512	General Diagnostic
			16384	Specific Event - Low Warning
			16416	Specific Event - Low Shutdown
			16448	Specific Event - High Warning
			16480	Specific Event - High Shutdown
			16512	Specific Event - High or Low Warning
			16544	Specific Event - High or Low Shutdown

SETPOINT VALUE LISTS

Block Number	Setpoint Number	Setpoint Name	Value	Value Name
97	1	Event Output Function #NN Trigger Condition	16576	Specific Event - Low Warning or Low Shutdown
			16608	Specific Event - Low Warning or Low Shutdown
			16640	Specific Event - Low Warning or Low Shutdown
			16672	Specific Event - High or Low Warning or High or Low Shutdown or Diagnostic
			16738	Specific Diagnostic - Data Erratic, Intermittent or Incorrect
			16739	Specific Diagnostic - Voltage Above Normal or Shorted to High Source
			16740	Specific Diagnostic - Voltage Below Normal or Shorted to Low Source
			16741	Specific Diagnostic - Current Below Normal or Open Circuit
			16742	Specific Diagnostic - Current Above Normal or Grounded Circuit
			16743	Specific Diagnostic - Mechanical System Not Responding or Out of Adjustment
			16744	Specific Diagnostic - Abnormal Frequency or Pulse Width or Period
			16745	Specific Diagnostic - Abnormal Update Rate
			16746	Specific Diagnostic - Abnormal Rate of Change
			16747	Specific Diagnostic - Root Cause Not Known
			16748	Specific Diagnostic - Bad Intelligent Device or Component
			16749	Specific Diagnostic - Out of Calibration
			16750	Specific Diagnostic - Special Instructions
			16755	Specific Diagnostic - Received Network Data In Error
			16768	Specific Event - Condition Exists
	2	Event Output Function #NN Suspect Parameter Number	38	External Tank Fuel Level
			82	Starting Air Pressure
			95	Fuel Filter Differential Pressure
			96	Fuel Level
			98	Engine Oil Level
			99	Engine Oil Filter Differential Pressure
			100	Engine Oil Pressure
			107	Air Filter Differential Pressure
			110	Engine Coolant Temperature
			111	Engine Coolant Level
			137	Fire Extinguisher Pressure
			167	Battery Charging System Voltage
			168	Battery Voltage

SETPOINT VALUE LISTS

Block Number	Setpoint Number	Setpoint Name	Value	Value Name
	2	Event Output Function #NN Suspect Parameter Number	171	Ambient Air Temperature
			173	Exhaust Temperature
			175	Engine Oil Temperature
			190	Engine Speed
			625	SCADA Data Link
			639	Primary Data Link
			701	Custom Event #1
			702	Custom Event #2
			703	Custom Event #3
			704	Custom Event #4
			705	Custom Event #5
			706	Custom Event #6
			707	Custom Event #7
			708	Custom Event #8
			709	Custom Event #9
			710	Custom Event #10
			711	Custom Event #11
			712	Custom Event #12
			713	Custom Event #13
			714	Custom Event #14
			715	Custom Event #15
			716	Custom Event #16
			924	Digital Output #1
			925	Digital Output #2
			926	Digital Output #3
			970	Auxiliary Engine Shutdown Switch
			1122	Generating Set Rear Bearing Temperature
			1231	Accessory Data Link
			1237	Emergency Shutdown Override Switch
			1383	Unexpected Engine Shutdown
			1390	Gas Pressure
			1664	Engine Failure to Start
			2433	Right Exhaust Temperature
			2434	Light Exhaust Temperature
			2436	Generating Set Output Frequency
			2440	Generating Set Output Voltage
			2448	Generating Set Current (Amp)
			2452	Generating Set Output Power
			2646	Digital Output #4
			2648	Maintenance Lamp
			3543	Engine Controller
			4000	Air Damper Closed
			4001	ATS in Normal Position

SETPOINT VALUE LISTS

Block Number	Setpoint Number	Setpoint Name	Value	Value Name
			4002	ATS in Emergency Position
			4003	Battery Charger Failure
			4004	Generating Set Circuit Breaker Closed
			4005	Utility Breaker Closed
			4006	Engine in Cooldown
			4007	Generating Set Control Not in Automatic
			4009	Generating Set Circuit Breaker Failure to Open
			4010	Utility Breaker Failure to Open
			4011	Generating Set Circuit Breaker Failure to Close
			4012	Utility Breaker Failure to Close
			4013	Generating Set Circuit Breaker Open
			4014	Utility Breaker Open
			4015	Utility to Generating Set Transfer Failure
			4016	Generating Set to Utility Transfer Failure
			4017	Loss of Utility
			4018	Generating Set Circuit Breaker Locked Out
			4019	Utility Breaker Locked Out
			4028	Earth Fault
			4029	Earth Leakage
104	1	Digital Selector #NN Source Configuration	0	Disabled
			1	Use Input #1
			2	Use Input #2
			3	Use Input #3
			4	Use Input #4
			5	Use Input #5
			6	Use Input #6
			7	Use Input #7
			8	Use Input #8
			9	Use Input #9
			10	Use Input #10
			11	Data Link
105	1	Event Input Function #NN Active State Configuration	0	Low
			1	High
	3	Event Input Function #NN Suspect Parameter Number	38	External Tank Fuel Level
			82	Starting Air Pressure
			95	Fuel Filter Differential Pressure
			96	Fuel Level
			98	Engine Oil Level
			99	Engine Oil Filter Differential Pressure
			100	Engine Oil Pressure
			107	Air Filter Differential Pressure
			110	Engine Coolant Temperature
			111	Engine Coolant Level

SETPOINT VALUE LISTS

Block Number	Setpoint Number	Setpoint Name	Value	Value Name
	3	Event Input Function #NN Suspect Parameter Number	137	Fire Extinguisher Pressure
			167	Battery Charging System Voltage
			168	Battery Voltage
			171	Ambient Air Temperature
			173	Exhaust Temperature
			175	Engine Oil Temperature
			701	Custom Event (Instance #1)
			702	Custom Event (Instance #2)
			703	Custom Event (Instance #3)
			704	Custom Event (Instance #4)
			705	Custom Event (Instance #5)
			706	Custom Event (Instance #6)
			707	Custom Event (Instance #7)
			708	Custom Event (Instance #8)
			709	Custom Event (Instance #9)
			710	Custom Event (Instance #10)
			711	Custom Event (Instance #11)
			712	Custom Event (Instance #12)
			713	Custom Event (Instance #13)
			714	Custom Event (Instance #14)
			715	Custom Event (Instance #15)
			716	Custom Event (Instance #16)
			1122	Generating Set Rear Bearing Temperature
			1239	Fuel Tank Leak
			1390	Gas Pressure
			2433	Right Exhaust Temperature
			2434	Left Exhaust Temperature
			2436	Generating Set Frequency
			2440	Generating Set Voltage
			2448	Generating Set Current
			2452	Generating Set Power
			4000	Air Damper Closed
			4001	ATS in Normal Position
			4002	ATS in Emergency Position
			4003	Battery Charger Failure
			4004	Generating Set Circuit Breaker Closed
			4005	Utility Circuit Breaker Closed
			4013	Generating Set Circuit Breaker Open
			4014	Utility Circuit Breaker Open
			4017	Loss of Utility
			4028	Earth Fault
			4029	Earth Leakage

SETPOINT VALUE LISTS

Block Number	Setpoint Number	Setpoint Name	Value	Value Name
106	4	Event Input Function #NN Failure Mode Identifier	0	High Shutdown
			1	Low Shutdown
			15	High Warning
			17	Low Warning
			31	Status
	1	Spare Analog Input Enable Status	0	Disabled
			1	Enabled
	2	Spare Analog Input Type Configuration	0	Pressure
			1	Temperature
			2	Level
109	3	Spare Analog Input Suspect Parameter Number	38	External Tank Fuel Level
			82	Starting Air Pressure
			95	Fuel Filter Differential Pressure
			96	Fuel Level
			98	Engine Oil Level
			99	Oil Filter Differential Pressure
			107	Air Filter Differential Pressure
			111	Engine Coolant Level
			137	Fire Extinguisher Pressure
			171	Ambient Air Temperature
			173	Exhaust Temperature
			175	Engine Oil Temperature
			1122	Generating Set Rear Bearing Temperature
109	1	Generating Set Winding Temperature Sensor Installation Status	0	Not Installed
			1	Installed
	2	Generating Set Bearing Temperature Sensor Installation Configuration	0	Not Installed
			1	Rear
			2	Front & Rear

BitMask Value Lists

Some setpoints in PowerWizard can take on multiple values simultaneously. This is particularly true for event response configurations, where an event can trigger multiple responses, such as a shutdown as well as a breaker tripping:

RESPONSE CONFIGURATION OPTIONS
Warning
Audible Alert
Soft Shutdown
Hard Shutdown
Utility Failure
Breaker #1 Trip
Breaker #2 Trip
Fault Protection Timer Enabled
Active Only